VOPĚNKA'S PRINCIPLE WITHOUT CHOICE: PRESERVATION UNDER SYMMETRIC EXTENSIONS

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Abstract. We give a detailed proof that every set-sized symmetric extension of a model of $\mathsf{ZF}+\mathsf{VP}$ again satisfies $\mathsf{ZF}+\mathsf{VP}$, where VP is formulated for arbitrary set-sized languages. For each standard $n \geq 1$, the same construction preserves $\mathsf{VP}(\Pi_n)$, allowing set parameters and arbitrary set-sized languages. The theories $\mathsf{ZF}+\mathsf{VP}$ and $\mathsf{ZFC}+\mathsf{VP}$ are equiconsistent, and $\mathsf{ZF}+\mathsf{VP}$ proves a proper class of Löwenheim–Skolem cardinals. Relative to $\mathrm{Con}(\mathsf{ZF}+\mathsf{VP})$, the axiom DC is independent of $\mathsf{ZF}+\mathsf{VP}+\neg\mathsf{AC}$, and there are Feferman–Lévy and full Solovay models satisfying VP. We consider whether every countable model of $\mathsf{ZF}+\mathsf{VP}$ has a class-generic extension satisfying $\mathsf{ZFC}+\mathsf{VP}$.

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1. Introduction

Vopěnka's principle asserts that every proper class of structures in a common set-sized language contains distinct members M and N with an elementary embedding $j : M \to N$. Throughout, VP denotes the corresponding first-order scheme for classes definable with set parameters. For each standard $n \geq 1$, $\mathsf{VP}(\Pi_n)$ restricts the defining formula to Π_n.

The equiconsistency in Theorem B is an unpublished result of Woodin that follows from his work in *Suitable extender models I* [22]. We give a detailed proof in Section 5.

Over ZFC, Bagaria showed that VP is equivalent to the existence of a $C^{(n)}$-extendible cardinal for every n [1]. Tzouvaras proved that a general downward-absoluteness principle for VP implies AC and asked whether AC is independent of $\mathsf{ZF}+\mathsf{VP}$ [20, Question 4.1]. The following two theorems answer this question.

Theorem A. *Every set-sized symmetric extension of a model of* $\mathsf{ZF}+\mathsf{VP}$ *satisfies* $\mathsf{ZF}+\mathsf{VP}$*. For every standard* $n \geq 1$*, every set-sized symmetric extension of a model of* $\mathsf{ZF}+\mathsf{VP}(\Pi_n)$ *satisfies* $\mathsf{ZF}+\mathsf{VP}(\Pi_n)$*.*

Theorem B (Woodin)**.** *The theories* $\mathsf{ZF}+\mathsf{VP}$ *and* $\mathsf{ZFC}+\mathsf{VP}$ *are equiconsistent.*

Applied to the basic Cohen symmetric system, Theorem A shows that $\mathrm{Con}(\mathsf{ZF}+\mathsf{VP})$ implies $\mathrm{Con}(\mathsf{ZF}+\mathsf{VP}+\neg\mathsf{AC})$, while Theorem B shows that $\mathrm{Con}(\mathsf{ZF}+\mathsf{VP})$ implies $\mathrm{Con}(\mathsf{ZFC}+\mathsf{VP})$. Hence AC is independent of $\mathsf{ZF}+\mathsf{VP}$, relative to $\mathrm{Con}(\mathsf{ZF}+\mathsf{VP})$.

Brooke-Taylor proved over ZFC that set forcing preserves VP for structures in a fixed finite language [5, Lemma 23 and Theorem 24]. Hayut and Karagila developed criteria for lifting elementary embeddings through symmetric extensions [9, Theorems 4.7 and 4.9]. Our proof constructs in the symmetric extension the graph of the restriction to the two structures.

The proof uses the set of all names of least possible rank. Applying VP to rank structures with these sets as distinguished constants gives an embedding that fixes the symmetric system. Proposition 2.4 then places the required restriction of its lift in the symmetric extension.

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For Theorem B, we first obtain a model with unbounded $C^{(n)}$-extendible cardinals for every standard n. Woodin's forcing at a sufficiently correct choiceless supercompact cardinal gives a ZFC rank model satisfying any prescribed finite subset of $\mathsf{ZFC}+\mathsf{VP}$. Compactness finishes the proof.

Sections 2 and 3 prove the lifting and preservation theorems. Section 4 derives a proper class of Löwenheim–Skolem cardinals, and Section 5 proves Theorem B. Sections 6–8 apply Theorem A to the Cohen, Feferman–Lévy, and full Solovay constructions. Section 9 asks whether Choice can instead be restored by class forcing over a given countable model while preserving VP.

The supplementary material [23] contains the supporting construction arguments and a correspondence with the Lean formalization.

All forcing notions have a largest condition $1_{\mathbb{P}}$. We write $\mathrm{rank}_{\mathrm{name}}(\tau)$ for the ground-model rank of a forcing name τ and $f\text{“}X$ for the direct image of X under f. We code structures so that the universe of a structure is contained in its transitive closure.

2. Symmetric models and the lifting lemma

Let $\mathscr{S}=\langle \mathbb{P},\Gamma,\mathcal{F}\rangle$ be a symmetric system, where Γ is a group of automorphisms of $\mathbb{P}$ and $\mathcal{F}$ is a normal filter of subgroups of Γ. We write πp for the action of π on a condition p and $\pi\cdot\sigma$ for the induced recursive action on a name σ. The latter is defined by $\pi\cdot\sigma=\{\langle \pi\cdot\tau,\pi p\rangle : \langle \tau,p\rangle\in\sigma\}$. Put $\mathrm{sym}(\sigma)=\{\pi\in\Gamma : \pi\cdot\sigma=\sigma\}$. A name is symmetric if its stabilizer belongs to $\mathcal{F}$. For a $\mathbb{P}$-name σ, let $\mathrm{cl}(\sigma)$ be the least set of names X with $\sigma\in X$ such that $\upsilon\in X$ whenever $\langle \upsilon,p\rangle\in\tau$ for some $\tau\in X$. A name σ is hereditarily symmetric if

$$\mathrm{sym}(\tau)\in\mathcal{F}\quad\text{for every }\tau\in\mathrm{cl}(\sigma). \tag{1}$$

Let HS denote the class of hereditarily symmetric names. Since $\sigma\in\mathrm{cl}(\sigma)$, every hereditarily symmetric name is symmetric. We write $\Vdash_{\mathrm{sym}}$ for the symmetric forcing relation. We use its definability, symmetry, and truth lemmas; see [12, Chapter 5], [13, Section 2], and [10, Lemma 2.2 and Theorem 2.5]. The action on names and the operation $\sigma\mapsto\mathrm{cl}(\sigma)$ are uniformly first-order definable by well-founded recursion.

We shall repeatedly use the following restriction lemma. For a transitive set a, write $\mathrm{Sat}(a)$ for the canonical satisfaction relation of $(a,\in)$.

Lemma 2.1. *Let $j:M\to N$ be elementary between transitive sets, and let $a\in M$ be transitive. If $\mathrm{Sat}(a)\in M$, then $j\restriction a:(a,\in)\longrightarrow(j(a),\in)$ is elementary.*

Proof. Relativizing each quantifier of φ to a, transitivity and elementarity of j give, for every formula φ and every finite tuple $\bar{x}$ from a,

$$(a,\in)\models\varphi(\bar{x})\quad\text{if and only if}\quad(j(a),\in)\models\varphi(j(\bar{x})).$$ □

The usual symmetric-model theorem holds over a ground model of ZF. The Power Set and Collection arguments are recorded because they require care without Choice.

Proposition 2.2. *Suppose that $V\models\mathsf{ZF}$, that $\mathscr{S}$ is set-sized, and that $G\subseteq\mathbb{P}$ is V-generic. Then $W=\mathsf{HS}^G=\{\sigma^G:\sigma\in\mathsf{HS}\}$ is a transitive model of ZF with $V\subseteq W\subseteq V[G]$.*

Proof. For $\sigma,\tau\in\mathsf{HS}$, define the intersection name

$$\tau\cap^\bullet\sigma=\{\langle \rho,r\rangle : \exists s\,(\langle \rho,s\rangle\in\sigma, r\le s,\text{ and } r\Vdash_{\mathrm{sym}}\rho\in\tau)\}.$$

Every $\pi\in\mathrm{sym}(\sigma)\cap\mathrm{sym}(\tau)$ fixes $\tau\cap^\bullet\sigma$, and every name appearing as the first coordinate of a pair in this name also appears as the first coordinate of a pair in σ. Hence $\tau\cap^\bullet\sigma\in\mathsf{HS}$. Moreover, $\tau\cap^\bullet\sigma\subseteq\mathrm{dom}(\sigma)\times\mathbb{P}$ and $(\tau\cap^\bullet\sigma)^G=\tau^G\cap\sigma^G$. Let $B_\sigma=\mathsf{HS}\cap\mathcal{P}(\mathrm{dom}(\sigma)\times\mathbb{P})$, which is a set by Separation, and define $\dot{Q}_\sigma=\{\langle u,p\rangle : u\in B_\sigma\text{ and }p\Vdash_{\mathrm{sym}}u\subseteq\sigma\}$. If $\pi\in\mathrm{sym}(\sigma)$, then π permutes B_σ, because it preserves $\mathrm{dom}(\sigma)\times\mathbb{P}$ and hereditary symmetry. For $u\in B_\sigma$ and $p\in\mathbb{P}$,

$$p\Vdash_{\mathrm{sym}}u\subseteq\sigma\quad\text{if and only if}\quad\pi p\Vdash_{\mathrm{sym}}\pi\cdot u\subseteq\sigma.$$

Thus $\mathrm{sym}(\sigma)\le\mathrm{sym}(\dot{Q}_\sigma)$. Since $\mathrm{sym}(\sigma)\in\mathcal{F}$ and every name occurring directly in $\dot{Q}_\sigma$ belongs to $B_\sigma\subseteq\mathsf{HS}$, we have $\dot{Q}_\sigma\in\mathsf{HS}$.

If $x \in \dot{Q}_\sigma^G$, then $x \in W$ and $x \subseteq \sigma^G$. Conversely, let $x \in W$ with $x \subseteq \sigma^G$, and choose $\tau \in \mathsf{HS}$ with $\tau^G = x$. Then $\tau \cap^\bullet \sigma \in B_\sigma$ and $(\tau \cap^\bullet \sigma)^G = x$. Since $(\tau \cap^\bullet \sigma)^G \subseteq \sigma^G$, there is $p \in G$ such that $p \Vdash_{\mathrm{sym}} \tau \cap^\bullet \sigma \subseteq \sigma$. Thus $x \in \dot{Q}_\sigma^G$, and consequently $\dot{Q}_\sigma^G = \mathcal{P}(\sigma^G) \cap W$. This proves Power Set in W.

To verify Collection, fix $\sigma, \vec{u} \in \mathsf{HS}$ and suppose $W \models \forall x \in \sigma^G \, \exists y \, \varphi(x, y, \vec{u}^G)$. Choose $p \in G$ such that $p \Vdash_{\mathrm{sym}} \forall x \in \sigma \, \exists y \, \varphi(x, y, \vec{u})$. Let $I = \{\langle \rho, s, r\rangle : \langle \rho, s\rangle \in \sigma, \ r \leq p, \text{ and } r \leq s\}$. If $\langle \rho, s, r\rangle \in I$, then $r \Vdash_{\mathrm{sym}} \rho \in \sigma$, so $r \Vdash_{\mathrm{sym}} \exists y \, \varphi(\rho, y, \vec{u})$. There are therefore $t \leq r$ and $\tau \in \mathsf{HS}$ such that $t \Vdash_{\mathrm{sym}} \varphi(\rho, \tau, \vec{u})$. Collection and Separation in V give a set C of quintuples $\langle \rho, s, r, t, \tau\rangle$ such that every $\langle \rho, s, r\rangle \in I$ occurs as the first three coordinates of a member of C, with $t \leq r$ and $t \Vdash_{\mathrm{sym}} \varphi(\rho, \tau, \vec{u})$.

Let $H = \mathrm{sym}(\sigma) \cap \bigcap_i \mathrm{sym}(u_i) \in \mathcal{F}$, put $b_0 = \{\langle \tau, t\rangle : \exists \rho, s, r \, \langle \rho, s, r, t, \tau\rangle \in C\}$, and set $b = \bigcup_{\pi \in H} \pi \cdot b_0$. Replacement shows that b is a set. Every $\xi \in H$ fixes b, since left multiplication by ξ permutes H. Thus $H \leq \mathrm{sym}(b)$. Each name occurring directly in b has the form $\pi \cdot \tau$ with $\tau \in \mathsf{HS}$, and normality of $\mathcal{F}$ gives $\pi \cdot \tau \in \mathsf{HS}$. Since $H \in \mathcal{F}$, we have $b \in \mathsf{HS}$. The identity belongs to H, so $b_0 \subseteq b$.

Let $x \in \sigma^G$. Choose $\langle \rho, s\rangle \in \sigma$ with $s \in G$ and $\rho^G = x$, and choose $r_0 \in G$ with $r_0 \leq p, s$. Define $D_{\rho,s} = \{t : \exists r, \tau \, \langle \rho, s, r, t, \tau\rangle \in C\}$. If $q \leq r_0$, then $\langle \rho, s, q\rangle \in I$, so C contains some $\langle \rho, s, q, t, \tau\rangle$ with $t \leq q$. Hence $D_{\rho,s}$ is dense below $r_0 \in G$, and genericity gives $t \in G \cap D_{\rho,s}$. Choose $\langle \rho, s, r, t, \tau\rangle \in C$. Then $\langle \tau, t\rangle \in b_0 \subseteq b$, so $\tau^G \in b^G$ and $W \models \varphi(x, \tau^G, \vec{u}^G)$.

Power Set and Collection are now established. For the remaining axioms, transitivity gives Extensionality and Foundation, canonical names give Empty Set, Pairing, Union, and Infinity, and symmetric subnames give Separation. Collection and Separation imply Replacement.

Thus $W \models \mathsf{ZF}$. □

Lemma 2.3. *Suppose that $V_\theta \models \mathsf{ZF}$ and $\mathrm{tc}(\{\mathscr{S}\}) \subseteq V_\theta$. If $\sigma \in V_\theta$ is a $\mathbb{P}$-name, then hereditary symmetry computed in V_θ agrees with hereditary symmetry computed in V. Consequently, if $V_\eta \models \mathsf{ZF}$ and $j : V_\theta \to V_\eta$ is elementary with $j(\mathscr{S}) = \mathscr{S}$, then $j(\sigma) \in \mathsf{HS}$ whenever $\sigma \in \mathsf{HS} \cap V_\theta$.*

Proof. Well-founded recursion is absolute to the transitive model V_θ, so the action on names and $\mathrm{cl}(\tau)$ are computed correctly there. Thus $\mathrm{cl}(\tau) \in V_\theta$. For every $\nu \in \mathrm{cl}(\tau)$, V_θ and V compute $\mathrm{sym}(\nu)$ from the same group Γ and agree whether it belongs to $\mathcal{F}$. Hence they agree on (1).

Now suppose that $j : V_\theta \to V_\eta$ is elementary, $j(\mathscr{S}) = \mathscr{S}$, and $\sigma \in \mathsf{HS} \cap V_\theta$. By the first part, V_θ regards σ as hereditarily symmetric, so elementarity gives that V_η regards $j(\sigma)$ as hereditarily symmetric with respect to $\mathscr{S}$.

Since $j(\mathscr{S}) = \mathscr{S} \in V_\eta$ and V_η is transitive, $\mathrm{tc}(\{\mathscr{S}\}) \subseteq V_\eta$. Applying the first part to V_η yields $j(\sigma) \in \mathsf{HS}$ in V. □

For names σ and τ, let $\langle \sigma, \tau\rangle^\bullet$ be the canonical name for their Kuratowski ordered pair. The construction is equivariant:

$$\pi \cdot \langle \sigma, \tau\rangle^\bullet = \langle \pi \cdot \sigma, \pi \cdot \tau\rangle^\bullet.$$

The canonical ordered-pair construction also preserves hereditary symmetry: if $\sigma, \tau \in \mathsf{HS}$, every new name in its Kuratowski code is fixed by $\mathrm{sym}(\sigma) \cap \mathrm{sym}(\tau)$; hence $\langle \sigma, \tau\rangle^\bullet \in \mathsf{HS}$.

Proposition 2.4. *Suppose that $V_\theta, V_\eta \models \mathsf{ZF}$, and let $j : V_\theta \to V_\eta$ be elementary. Suppose that $j(z) = z$ for every $z \in \mathrm{tc}(\{\mathscr{S}\})$. If $G \subseteq \mathbb{P}$ is V-generic and $W = \mathsf{HS}^G$, then there is a function $E_j \in W$ such that*

$$E_j(\sigma^G) = j(\sigma)^G \qquad (\sigma \in \mathsf{HS} \cap V_\theta).$$

Thus, in $V[G]$, $E_j = j_G \restriction \{\sigma^G : \sigma \in \mathsf{HS} \cap V_\theta\}$, where $j_G : V_\theta[G] \to V_\eta[G]$ is the ordinary forcing lift of j, defined by $j_G(\sigma^G) = j(\sigma)^G$.

Proof. Since j fixes every member of $\mathrm{tc}(\{\mathscr{S}\})$, it fixes $\mathbb{P}$ pointwise, so $j``G = G$. Since G is V-generic, it is generic over both V_θ and V_η.

Define $j_G(\sigma^G) = j(\sigma)^G$ for every $\mathbb{P}$-name $\sigma \in V_\theta$. The forcing theorem and elementarity show that this is well defined and that

$$p \Vdash_{V_\theta} \varphi(\vec{\sigma}) \quad \text{if and only if} \quad p \Vdash_{V_\eta} \varphi(j(\vec{\sigma}))$$

for every formula φ, condition $p \in \mathbb{P}$, and finite tuple $\vec{\sigma}$ of names from V_θ. Thus $j_G : V_\theta[G] \to V_\eta[G]$ is elementary. Put $D_\theta = \mathsf{HS} \cap V_\theta$. Hereditary symmetry is first-order definable from $\mathscr{S}$, so D_θ is a set by Separation. Lemma 2.3 gives $j(\sigma) \in \mathsf{HS}$ for every $\sigma \in D_\theta$.

The action of Γ on $\mathbb{P}$-names is defined by well-founded recursion. Hence, for $\pi \in \Gamma$ and every $\mathbb{P}$-name $\sigma \in V_\theta$, elementarity and $j(\pi) = \pi$ give

$$j(\pi \cdot \sigma) = j(\pi) \cdot j(\sigma) = \pi \cdot j(\sigma). \tag{2}$$

Moreover, $\mathrm{cl}(\pi \cdot \sigma) = \pi\text{“}\,\mathrm{cl}(\sigma)$, and $\mathrm{sym}(\pi \cdot \nu) = \pi\,\mathrm{sym}(\nu)\pi^{-1}$. Since $\mathcal{F}$ is normal, σ is hereditarily symmetric if and only if $\pi \cdot \sigma$ is hereditarily symmetric. As π bijects the $\mathbb{P}$-names in V_θ, it follows that $\pi\text{“}D_\theta = D_\theta$.

The required graph name is

$$\dot{E}_j = \{\langle\langle \sigma, j(\sigma)\rangle^\bullet, 1_\mathbb{P}\rangle : \sigma \in D_\theta\}.$$

Since D_θ is a set, $\dot{E}_j$ is a $\mathbb{P}$-name, and every pair name occurring in it belongs to HS. For each $\pi \in \Gamma$, equivariance of the ordered-pair name, $\pi 1_\mathbb{P} = 1_\mathbb{P}$, equation (2), and $\pi\text{“}D_\theta = D_\theta$ give

$$\pi \cdot \dot{E}_j = \{\langle\langle \pi \cdot \sigma, j(\pi \cdot \sigma)\rangle^\bullet, 1_\mathbb{P}\rangle : \sigma \in D_\theta\} = \dot{E}_j.$$

Thus $\mathrm{sym}(\dot{E}_j) = \Gamma \in \mathcal{F}$, so $\dot{E}_j \in \mathsf{HS}$.

Let $E_j = \dot{E}_j^G$. Then

$$E_j = \{\langle \sigma^G, j(\sigma)^G\rangle : \sigma \in D_\theta\}.$$

If $\sigma^G = \tau^G$, the well-definedness of j_G gives $j(\sigma)^G = j(\tau)^G$. Hence E_j is a function in W, with

$$E_j(\sigma^G) = j(\sigma)^G \qquad (\sigma \in D_\theta). \qquad \square$$

3. Preservation

Lemma 3.1. *Let $\theta > \omega$ be a limit ordinal. If no function $f : a \to \theta$ with $a \in V_\theta$ has cofinal range, then $V_\theta \models \mathsf{ZF}$. The hypothesis on θ is Π_1.*

Proof. Only Replacement and Collection require proof. Suppose $a, p \in V_\theta$ and $V_\theta \models \forall x \in a\, \exists! y\, \varphi(x, y, p)$. Replacement in V gives the corresponding function F. If the ranks of its values were unbounded in θ, then $x \mapsto \mathrm{rank}(F(x))$ would be cofinal in θ, contrary to the hypothesis. Hence $F\text{“}a \in V_\theta$.

The same rank argument proves Collection. Suppose $V_\theta \models \forall x \in a\, \exists y\, \varphi(x, y, p)$. For each $x \in a$, take the least rank of a suitable $y \in V_\theta$. These ranks are bounded below θ. If $\beta < \theta$ lies above them, Separation applied to V_β gives a set containing a suitable y for every $x \in a$. Finally, $x \in V_\theta$ is Δ_1 in θ, and "$f : a \to \theta$ is cofinal" is Δ_0 in f, a, θ. Thus the hypothesis is Π_1. $\square$

Lemma 3.2. *Assume $\mathsf{ZF} + \mathsf{VP}(\Pi_1)$. Above every ordinal γ there is a limit ordinal κ such that no map from any $a \in V_\kappa$ is cofinal in κ. Consequently $V_\kappa \models \mathsf{ZF}$. Under AC, such a κ may be chosen measurable.*

Proof. Fix γ and put $\bar{\gamma} = \max\{\gamma, \omega\}$. For $\lambda > \bar{\gamma}$, consider

$$\mathcal{B}_\lambda = \langle V_{\lambda+\omega}, \in, \lambda, (z)_{z \in V_{\bar{\gamma}+1}}\rangle$$

in the set-sized language with membership, one distinguished constant, and constants for $V_{\bar{\gamma}+1}$. The structures $\mathcal{B}_\lambda$ form a Π_1-definable proper class: the hierarchy equation is Π_1, ordinal addition is Δ_1, and the interpretation tables are bounded in the code. Applying $\mathsf{VP}(\Pi_1)$ to this class gives distinct ordinals λ, μ and an elementary embedding $j : \mathcal{B}_\lambda \longrightarrow \mathcal{B}_\mu$. By elementarity and the choice of constants, $j(\lambda) = \mu$, while $j(z) = z$ for every $z \in V_{\bar{\gamma}+1}$. Since $\lambda \neq \mu$, the embedding moves an ordinal. Let κ be the least ordinal moved by j. Then $\bar{\gamma} < \kappa \leq \lambda$, $j(\kappa) > \kappa$, and a rank induction shows that $j(x) = x$ for every $x \in V_\kappa$.

Suppose that $f : a \to \kappa$ is cofinal for some $a \in V_\kappa$. The embedding fixes a and each of its elements, so $j(f) : a \to j(\kappa)$ is cofinal and $j(f)(x) = f(x) < \kappa$ for every $x \in a$. This is impossible because $j(\kappa) > \kappa$. Thus no map from a member of V_κ is cofinal in κ. The ordinal κ cannot be a successor, since j fixes every smaller ordinal. As $\kappa > \bar{\gamma} \geq \omega$, Lemma 3.1 applies, so $V_\kappa \models \mathsf{ZF}$.

Assume now that AC holds and define $U = \{X \subseteq \kappa : \kappa \in j(X)\}$.

Elementarity shows that U is an ultrafilter on κ. Because j fixes every ordinal below κ, U is nonprincipal. Moreover, U is κ-complete, since $V_{\lambda+\omega}$ contains every sequence of fewer than κ subsets of κ. Under Choice, κ must be a cardinal. If it were not, then κ could be partitioned into fewer than κ singletons, contradicting the κ-completeness of the nonprincipal ultrafilter U. Thus U is a nonprincipal κ-complete ultrafilter on the cardinal κ, and therefore κ is measurable. □

Theorem 3.3. *Let $V \models \mathsf{ZF} + \mathsf{VP}$, with VP understood in the arbitrary-language sense fixed in the Introduction. Let $\mathscr{S} = \langle \mathbb{P}, \Gamma, \mathcal{F} \rangle \in V$ be a set-sized symmetric system, and let $G \subseteq \mathbb{P}$ be V-generic. Then*

$$W = \mathsf{HS}^G \models \mathsf{ZF} + \mathsf{VP}.$$

Proof. Proposition 2.2 gives $W \models \mathsf{ZF}$. Fix a formula $\varphi(x, y)$, a parameter $a \in W$, and a set-sized language $L \in W$ such that $\mathcal{A} = \{M \in W : W \models \varphi(M, a)\}$ is a proper class of L-structures. Choose $\dot{a}, \dot{L} \in \mathsf{HS}$ with $\dot{a}^G = a$ and $\dot{L}^G = L$. For $p \in \mathbb{P}$, let R_p be the class of ordinals α for which there is a name $\tau \in \mathsf{HS}$ satisfying

$$p \Vdash_{\mathrm{sym}} \text{“}\tau \text{ is an } \dot{L}\text{-structure, } \varphi(\tau, \dot{a}), \text{ and } \operatorname{rank}(\tau) = \check{\alpha}.\text{”} \tag{3}$$

In (3), $\operatorname{rank}(\tau)$ denotes the rank of the value of τ in the generic extension. By contrast, $\operatorname{rank}_{\mathrm{name}}(\tau)$ is the von Neumann rank of the name τ itself.

There is a condition $q \in G$ for which R_q is unbounded in Ord. Suppose not. Let $D_{\mathrm{bd}} = \{p \in \mathbb{P} : R_p \text{ is bounded}\}$ and, for $p \in D_{\mathrm{bd}}$, let $b(p)$ be the least ordinal such that $R_p \subseteq b(p)$. The domain D_{bd} is a subset of the set $\mathbb{P}$, and definability of symmetric forcing together with Replacement makes b a set function in V. Our supposition gives $G \subseteq D_{\mathrm{bd}}$.

In $V[G]$, set $\delta = \sup b\text{“}G$. This is an ordinal, so preservation of ordinals gives $\delta \in V \subseteq W$. If $M \in \mathcal{A}$, choose $\tau \in \mathsf{HS}$ with $\tau^G = M$ and let $\alpha = \operatorname{rank}(M)$. By the symmetric truth lemma, some $p \in G$ forces the assertion in (3); hence $\alpha < b(p) \leq \delta$. Thus $\mathcal{A} \subseteq V_\delta^W$, and Separation in W makes $\mathcal{A}$ a set, a contradiction.

For $\alpha \in R_q$, let $\Psi_\alpha(\tau)$ assert that $\tau \in \mathsf{HS}$ and that (3) holds with q in place of p. We use the following definitions:

$$\nu_\alpha = \min\{\operatorname{rank}_{\mathrm{name}}(\tau) : \Psi_\alpha(\tau)\}, \qquad X_\alpha = \{\tau \in V_{\nu_\alpha+1} : \operatorname{rank}_{\mathrm{name}}(\tau) = \nu_\alpha \text{ and } \Psi_\alpha(\tau)\}.$$

The ordinal ν_α exists, X_α is a nonempty set by Separation, and both are uniformly definable from α and the fixed parameters.

Choose $\rho > \omega$ such that $\operatorname{tc}(\{\mathscr{S}, \dot{a}, \dot{L}\}) \subseteq V_\rho$. By Lemma 3.2, for each $\alpha \in R_q$ there are arbitrarily high θ with $V_\theta \models \mathsf{ZF}$. The relevant rank is

$$\theta_\alpha = \min\{\theta : V_\theta \models \mathsf{ZF} \text{ and } \theta > \max\{\rho, \alpha, \operatorname{rank}(X_\alpha)\} + \omega\}.$$

In the common language consisting of a binary relation symbol, two distinguished constant symbols, and constant symbols c_z for $z \in V_\rho$, use the structure

$$\mathcal{M}_\alpha = \langle V_{\theta_\alpha}, \in, \alpha, X_\alpha, (z)_{z \in V_\rho} \rangle \qquad (\alpha \in R_q).$$

The structures $\mathcal{M}_\alpha$ form a definable proper class since R_q is unbounded. By VP, there is an elementary embedding $j : \mathcal{M}_\alpha \to \mathcal{M}_\beta$ with $\alpha \neq \beta$. The distinguished constants give $j(\alpha) = \beta$, $j(X_\alpha) = X_\beta$, and $j(z) = z$ for every $z \in V_\rho$.

Fix $\tau \in X_\alpha$. Then $j(\tau) \in X_\beta$, and the definitions of X_α and X_β give

$$q \Vdash_{\mathrm{sym}} \text{“}\tau \text{ and } j(\tau) \text{ are } \dot{L}\text{-structures satisfying } \varphi(-, \dot{a}) \text{ with ranks } \check{\alpha} \text{ and } \check{\beta}.\text{”} \tag{4}$$

Since j fixes V_ρ pointwise, it fixes every member of $\operatorname{tc}(\{\mathscr{S}, \dot{a}, \dot{L}\})$.

Apply Proposition 2.4 to obtain the elementary lift $j_G : V_{\theta_\alpha}[G] \to V_{\theta_\beta}[G]$ and a function $E_j \in W$ agreeing with j_G on values of names in $\mathsf{HS} \cap V_{\theta_\alpha}$. Put $M = \tau^G$ and $N = j(\tau)^G$. Then $j_G(M) = N$.

If $s \in L = \dot{L}^G$, then $s = \sigma^G$ for some $\langle \sigma, p \rangle \in \dot{L}$ with $p \in G$. Since $\sigma \in \operatorname{cl}(\dot{L}) \subseteq V_\rho$, we have $j(\sigma) = \sigma$, and hence $j_G(s) = s$. Thus j_G fixes L pointwise. Moreover, $j(\dot{L}) = \dot{L}$ and $j_G(L) = L$.

By the coding convention of the Introduction, $|M| \subseteq \operatorname{tc}(M)$. An induction on name evaluation shows that every element of $\operatorname{tc}(M)$ is σ^G for some $\sigma \in \operatorname{cl}(\tau)$. Since $\tau \in \mathsf{HS} \cap V_{\theta_\alpha}$, every such σ

belongs to $\mathsf{HS} \cap V_{\theta_\alpha}$. Hence $|M| \subseteq \mathrm{dom}(E_j)$, and $e = E_j \restriction |M|$ belongs to W. In $V[G]$, the map e is $j_G \restriction |M|$. Since the universe of a structure is uniformly definable from its fixed set code,

$$j_G(|M|) = |j_G(M)| = |N|, \qquad j_G\text{“}|M| \subseteq |N|.$$

Satisfaction for set-sized structures is absolute between the transitive models under consideration. Since j_G fixes L pointwise, elementarity gives

$$M \models \psi(\bar{m}) \quad \text{if and only if} \quad N \models \psi(e(\bar{m}))$$

for every L-formula ψ and every finite tuple $\bar{m}$ from $|M|$. Thus W regards e as an L-elementary embedding from M to N.

Finally, $q \in G$ and (4) imply that $M, N \in \mathcal{A}$, with ranks α and β. Thus $M \neq N$. This proves the arbitrary instance of VP, so $W \models \mathsf{ZF} + \mathsf{VP}$. □

Corollary 3.4. *Let $n \geq 1$ be standard. If $V \models \mathsf{ZF} + \mathsf{VP}(\Pi_n)$, $\mathscr{S} = \langle \mathbb{P}, \Gamma, \mathcal{F} \rangle \in V$ is a set-sized symmetric system, $G \subseteq \mathbb{P}$ is V-generic, and $W = \mathsf{HS}^G$, then*

$$W \models \mathsf{ZF} + \mathsf{VP}(\Pi_n).$$

Proof. For each fixed standard formula of positive Lévy level $k \geq 1$, symmetric forcing has a definition of the same polarity and level. The bounded base is Δ_1: existence and uniqueness of the name and forcing recursion tables give its two presentations. At an existential step, the density clause quantifies over an HS name; condition quantifiers are bounded by $\mathbb{P}$. Collection supplies the recursion certificates without selecting one name per condition, and negation gives the dual presentation.

We use the following elementary coding observation. If $S(z)$ is Σ_k, $k \geq 1$, write it as $\exists u\, \psi(z, u)$ with ψ in Π_{k-1}. When $S(z)$ holds, let ξ be the least rank of such a u and let C be the set of all such u of rank ξ. There is a uniform Π_k formula $\chi_S(z, c)$ saying that

$$c = \langle \xi, V_\xi, V_{\xi+1}, C \rangle$$

has these properties. Indeed, once the two rank segments are included in c, all quantifiers used for minimality and for the exact description of C are bounded; the equations defining the rank segments are Π_1. Hence c is unique and $\chi_S(z, c)$ implies $S(z)$.

The Π_n argument uses this observation twice: to record the least name rank together with all witnesses at that rank, and to record the least rank satisfying Lemma 3.1.

Lemma 3.2 applies since $\mathsf{VP}(\Pi_n)$ includes $\mathsf{VP}(\Pi_1)$, and gives unboundedly many ordinals satisfying the Π_1 condition of Lemma 3.1. For $b < \theta$, the statement that no ordinal η with $b < \eta < \theta$ satisfies this condition is Σ_1. By the preceding observation, this statement can be expressed by a Π_1 formula involving an additional set d, uniquely determined by b and θ. Thus the pair (θ, d), where θ is the least such ordinal above b, is Π_1-definable from b. The satisfaction relation of V_θ lies in $V_{\theta+\omega}$.

Now let a Π_n formula $\varphi(x, a)$ define a proper class $\mathcal{A}$ of structures in a set-sized language L in W. Choose names $\dot{a}, \dot{L} \in \mathsf{HS}$. Define R_p as in the proof of Theorem 3.3. The same boundedness argument gives $q \in G$ for which R_q is unbounded. Let $\Psi(\alpha, \tau)$ say that $\tau \in \mathsf{HS}$ and that q forces τ to be an $\dot{L}$-structure of rank α satisfying $\varphi(\tau, \dot{a})$. By the first paragraph, Ψ is Π_n.

For $\alpha \in R_q$, let ν_α be the least von Neumann rank of a name τ satisfying $\Psi(\alpha, \tau)$. We use the following notation:

$$U_\alpha = V_{\nu_\alpha}, \qquad U_\alpha^+ = V_{\nu_\alpha+1}, \qquad X_\alpha = \{\tau \in U_\alpha^+ : \mathrm{rank}_{\mathrm{name}}(\tau) = \nu_\alpha \text{ and } \Psi(\alpha, \tau)\}.$$

The assertions

$$\forall \tau \in U_\alpha\, \neg\Psi(\alpha, \tau) \quad \text{and} \quad \forall \tau \in U_\alpha^+ \big((\mathrm{rank}_{\mathrm{name}}(\tau) = \nu_\alpha \wedge \Psi(\alpha, \tau)) \longrightarrow \tau \in X_\alpha\big)$$

are Σ_n. Apply the preceding coding observation separately to them, and let c_α be the ordered pair of the resulting codes. Its first component records that no name of rank below ν_α satisfies $\Psi(\alpha, \cdot)$; its second records that every name of rank ν_α satisfying $\Psi(\alpha, \cdot)$ belongs to X_α. Keeping c_α as a distinguished component lets the defining formula state both coded Π_n conditions without another existential quantifier. Together with the rank-segment equations and the assertion that

X_α is the nonempty set of rank-ν_α names satisfying $\Psi(\alpha,\cdot)$, these clauses give a Π_n definition from α of

$$\langle \nu_\alpha, U_\alpha, U_\alpha^+, X_\alpha, c_\alpha\rangle.$$

Choose $\rho > \omega$ with $\mathrm{tc}(\{\mathscr{S}, \dot a, \dot L\}) \subseteq V_\rho$. Let θ_α be the least limit ordinal above $\max\{\rho, \alpha, \mathrm{rank}(\langle \nu_\alpha, U_\alpha, U_\alpha^+, X_\alpha, c_\alpha\rangle)\} + \omega$ admitting no cofinal map $f : a \to \theta_\alpha$ with $a \in V_{\theta_\alpha}$, and let d_α be the unique code witnessing this leastness in the Π_1 definition described above. Put $h_\alpha = \max\{\theta_\alpha, \mathrm{rank}(d_\alpha)\} + \omega$. In the set-sized language with membership, finitely many distinguished constants, and constants for V_ρ, let

$$\mathcal{N}_\alpha = \langle V_{h_\alpha}, \in, \alpha, \nu_\alpha, U_\alpha, U_\alpha^+, X_\alpha, c_\alpha, \theta_\alpha, d_\alpha, (z)_{z\in V_\rho}\rangle.$$

The rank-segment equations and the definition of θ_α are Π_1, the least-name-rank clause is Π_n, and the other clauses, including the definition of h_α, are bounded or Δ_1. Thus $\{\mathcal{N}_\alpha : \alpha \in R_q\}$ is a Π_n-definable proper class: R_q is unbounded, and the tuple is unique for each α.

Apply $\mathsf{VP}(\Pi_n)$ in V. There are distinct $\mathcal{N}_\alpha, \mathcal{N}_\beta$ and an elementary embedding $j : \mathcal{N}_\alpha \to \mathcal{N}_\beta$. Uniqueness gives $\alpha \neq \beta$, and the distinguished constants give $j(\alpha) = \beta$, $j(X_\alpha) = X_\beta$, and $j(\theta_\alpha) = \theta_\beta$. Since $\mathrm{Sat}(V_{\theta_\alpha}) \in V_{h_\alpha}$, Lemma 2.1 shows that $j_0 = j \restriction V_{\theta_\alpha} : V_{\theta_\alpha} \longrightarrow V_{\theta_\beta}$ is elementary. Both ranks satisfy ZF, and the constants from V_ρ make j_0 fix the symmetric system, every condition, and the names $\dot a$ and $\dot L$.

Choose $\tau \in X_\alpha$. Then $j_0(\tau) \in X_\beta$, and q forces that the two names denote members of $\mathcal{A}$ of ranks α and β. Proposition 2.4 lifts j_0 through G. The restriction of the lifted map to the universe of τ^G lies in W. The lift fixes L pointwise, so this restriction is an elementary L-embedding from τ^G to $j_0(\tau)^G$. Since $q \in G$ and $\alpha \neq \beta$, these are distinct members of $\mathcal{A}$. □

This corollary, together with Theorem 3.3, proves Theorem A.

4. Löwenheim–Skolem cardinals

We use Usuba's definitions of weakly LS and LS cardinals [21, Definition 1.1].

Definition 4.1. Let κ be an uncountable cardinal.

(1) κ is *weakly Löwenheim–Skolem*, or *weakly LS*, if for every $\gamma < \kappa$, every $\alpha \geq \kappa$, and every $x \in V_\alpha$, there is $X \prec V_\alpha$ such that $V_\gamma \subseteq X$, $x \in X$, and the transitive collapse of X belongs to V_κ.

(2) κ is *Löwenheim–Skolem*, or *LS*, if for every $\gamma < \kappa$, every $\alpha \geq \kappa$, and every $x \in V_\alpha$, there are $\beta \geq \alpha$ and $X \prec V_\beta$ such that $V_\gamma \subseteq X$, $x \in X$, the transitive collapse of X belongs to V_κ, and

$$^{V_\gamma}(X \cap V_\alpha) \subseteq X.$$

The last clause means that every function from V_γ into $X \cap V_\alpha$ belongs to X.

Lemma 4.2. *A limit of LS cardinals is LS, and a limit of weakly LS cardinals is weakly LS.*

Proof. Suppose that κ is a limit of LS cardinals. It is uncountable, and it is a cardinal: if $b : \kappa \to \alpha$ were a bijection for some $\alpha < \kappa$, an LS cardinal μ with $\alpha < \mu < \kappa$ would make $b \restriction \mu$ an injection from the initial ordinal μ into the smaller ordinal α, a contradiction. Now fix $\gamma < \kappa \leq \alpha$ and $x \in V_\alpha$. Choose an LS cardinal μ with $\gamma < \mu < \kappa$, and apply its LS property to (γ, α, x). The submodel supplied by the LS property of μ has transitive collapse in $V_\mu \subseteq V_\kappa$, so it also witnesses the LS property for κ. The weakly LS argument is the same. □

For $n \geq 1$, put $C^{(n)} = \{\alpha : V_\alpha \prec_{\Sigma_n} V\}$.

Mohammd's characterizations [16, Theorems 3.1 and 2.8] imply the following theorem.

Theorem 4.3. *The theory* $\mathsf{ZF} + \mathsf{VP}$ *proves that there is a proper class of LS cardinals.*

Proof. VP includes $\mathsf{VP}(\Pi_2)$. By [16, Theorem 3.1], there is a proper class of 1-choiceless extendible ordinals. Each is a cardinal by the observation following [16, Definition 2.1], and [16, Theorem 2.8] identifies them with the 2-choiceless supercompact cardinals. It is therefore enough to prove that every 2-choiceless supercompact cardinal is LS.

Let κ be 2-choiceless supercompact, and fix $\delta < \kappa \leq \alpha$ and $x \in V_\alpha$. Choose $\lambda \in C^{(2)}$ above $\alpha + \omega$. Apply [16, Definition 2.3] with lower bound δ and parameter $a = \langle \alpha, x, \alpha + \omega \rangle$. We obtain $\bar{\lambda} < \kappa$ in $C^{(2)}$, an element $\bar{a} \in V_{\bar{\lambda}}$, and an elementary embedding $j : V_{\bar{\lambda}} \longrightarrow V_\lambda$ such that $\mathrm{crit}(j) > \delta$ and $j(\bar{a}) = a$. Decoding $\bar{a}$ gives ordinals $\bar{\alpha} < \bar{\lambda}$ and $\bar{x} \in V_{\bar{\alpha}}$ with $\bar{a} = \langle \bar{\alpha}, \bar{x}, \bar{\alpha} + \omega \rangle$, $j(\bar{\alpha}) = \alpha$, $j(\bar{x}) = x$, and $\bar{\alpha} + \omega < \bar{\lambda}$. Moreover $\bar{\alpha} > \delta$, because j fixes every ordinal below $\mathrm{crit}(j)$ whereas $j(\bar{\alpha}) = \alpha \geq \kappa > \delta$.

Put $X = j\text{“}V_{\bar{\lambda}}$. Then $X \prec V_\lambda$, $V_\delta \subseteq X$, $x \in X$, and the transitive collapse of X is $V_{\bar{\lambda}} \in V_\kappa$. Moreover, $X \cap V_\alpha = j\text{“}V_{\bar{\alpha}}$. If $f : V_\delta \to X \cap V_\alpha$, let $\bar{f}(u)$ be the unique $y \in V_{\bar{\alpha}}$ such that $j(y) = f(u)$. Replacement gives $\bar{f}$, and $\bar{\alpha} + \omega < \bar{\lambda}$ implies $\bar{f} \in V_{\bar{\lambda}}$. Since j fixes V_δ pointwise, $j(\bar{f}) = f$. Thus ${}^{V_\delta}(X \cap V_\alpha) \subseteq X$, and κ is LS. □

Corollary 4.4. *Above every ordinal there is a singular LS cardinal of cofinality ω.*

Proof. Starting above the given ordinal, recursively take the least LS cardinal above the preceding one and let κ be the supremum of the resulting strictly increasing ω-sequence. Lemma 4.2 makes κ LS, and the sequence witnesses $\mathrm{cf}(\kappa) = \omega$. □

Corollary 4.5. *The theory* $\mathsf{ZF} + \mathsf{VP}$ *proves that there is a definable class forcing which forces* ZFC.

Proof. Combine Theorem 4.3 with Usuba's Choice-restoration theorem [21, Corollary 4.8]; the resulting class forcing forces ZFC. □

5. Consistency transfer to Choice

Woodin traces the Choice-restoring arguments used here to his early work introducing $\mathsf{I0}$ and proving determinacy in $L(\mathbb{R})$. He needed to obtain $\mathsf{I0}$ in a universe satisfying Choice to establish it as a large-cardinal hypothesis over ZFC. Starting from ZF with a super-Reinhardt cardinal, he obtained an intermediate setting satisfying λ-DC with an elementary embedding $j : V \to V$, where $\lambda = \kappa_\omega(j) = \sup_{n<\omega} j^n(\mathrm{crit}(j))$. From this setting he then obtained a model of $\mathsf{ZFC} + \mathsf{I0}$.[1]

Proposition 5.4 reduces the forward consistency implication to a model with unbounded $C^{(n)}$-extendible cardinals for every standard n. If this model satisfies Choice, it already satisfies $\mathsf{ZFC} + \mathsf{VP}$. Otherwise, for each finite $F \subseteq \mathsf{ZFC} + \mathsf{VP}$, Theorem 5.11 produces a ZFC rank model satisfying F. Compactness then yields a model of $\mathsf{ZFC} + \mathsf{VP}$.

Choiceless extendibility. Following [16, Definition 6.6], let $E_n(\kappa)$ assert that κ is a cardinal and that, whenever $\lambda > \kappa$ belongs to $C^{(n)}$, there are $\theta > \lambda$ in $C^{(n)}$ and an elementary embedding $j : V_\lambda \to V_\theta$ with $\mathrm{crit}(j) = \kappa$ and $j(\kappa) > \lambda$. We call such a κ $C^{(n)}$-extendible; over ZFC, this agrees with Bagaria's notion. Let UE, for unbounded extendibility, be the recursive scheme asserting that the E_n-cardinals are unbounded for every $n \geq 1$. This fixed-critical-point notion differs from the variable-critical-point supercompactness used in Theorem 4.3.

Lemma 5.1. *Over* ZF, UE *implies* VP.

Proof. Let $\mathcal{A} = \{M : \varphi(M, a)\}$ be a proper class of structures in a set-sized language L, and put $\rho = \mathrm{rank}(\mathrm{tc}(\{a, L\}))$. Choose n large enough that $C^{(n)}$-correct ranks are correct about φ, the coding of L-structures, and satisfaction for set-sized L-structures. Choose a $C^{(n)}$-extendible cardinal $\kappa > \rho$ and $M \in \mathcal{A}$ with $\alpha = \mathrm{rank}(M) > \kappa$. Let $\lambda \in C^{(n)}$ lie above $\alpha + \omega$, and take $j : V_\lambda \longrightarrow V_\theta$ witnessing $C^{(n)}$-extendibility of κ, with $\theta \in C^{(n)}$. The parameters a and L are fixed. Correctness of the source and target ranks, followed by elementarity, gives $j(M) \in \mathcal{A}$, and $j \restriction |M| : M \to j(M)$ is L-elementary. Finally,

$$\mathrm{rank}(j(M)) = j(\alpha) \geq j(\kappa) > \lambda > \alpha,$$

so the two structures are distinct. □

[1] W. H. Woodin, personal communication, 7 October 2026.

We recall that a nonzero cardinal δ is rank-Berkeley if, for every $\zeta < \delta$ and every $\Theta > \delta$, there is a nontrivial elementary self-embedding $j : V_\Theta \to V_\Theta$ with $\zeta < \mathrm{crit}(j) < \delta$ and $j(\delta) = \delta$ [16, Definition 4.4].

Lemma 5.2. ZF *proves that the existence of a rank-Berkeley cardinal implies the existence of a set model of*

$$\mathsf{ZF} + \mathsf{UE} + \mathsf{VP}.$$

Proof. Let δ be rank-Berkeley. Taking $\zeta = 0 < \delta$, choose $\Theta > \delta + \omega$ and a nontrivial elementary embedding $j : V_\Theta \longrightarrow V_\Theta$ with critical point $\kappa_0 < \delta$ and $j(\delta) = \delta$. Put $\kappa_{m+1} = j(\kappa_m)$ and $\lambda = \sup_{m<\omega} \kappa_m$. Inductively, $\kappa_m < \delta$ for all m, so $\lambda \leq \delta < \Theta$. The sequence $s = \langle \kappa_m : m < \omega \rangle$ belongs to V_Θ, and $j(s)$ is its tail. Since $\lambda = \sup \mathrm{ran}(s)$ is a first-order relation between objects in the domain,

$$j(\lambda) = \sup \mathrm{ran}(j(s)) = \sup_{m<\omega} \kappa_{m+1} = \lambda.$$

Hence $j \restriction V_\lambda : V_\lambda \to V_\lambda$ is elementary.

If $f : a \to \kappa_0$ were cofinal for some $a \in V_{\kappa_0}$, then $j(f)$ would be cofinal in κ_1 although its range remained below κ_0. Lemma 3.1 therefore gives $V_{\kappa_0} \models \mathsf{ZF}$. Since $j \restriction V_{\kappa_0}$ is the identity, Lemma 2.1 gives $V_{\kappa_0} \prec V_{\kappa_1}$. Applying j^m to this assertion gives

$$V_{\kappa_m} \prec V_{\kappa_{m+1}} \qquad (m < \omega).$$

The elementary-chain theorem yields $V_{\kappa_m} \prec V_\lambda$ for every m, and $V_\lambda \models \mathsf{ZF}$.

Fix $n \geq 1$ and $i < \omega$. We show inside V_λ that κ_i is $C^{(n)}$-extendible. For $\mu < \lambda$, internal membership in $C^{(n)}$ is characterized by

$$\mu \in (C^{(n)})^{V_\lambda} \quad \text{if and only if} \quad V_\mu \prec_{\Sigma_n} V_{\kappa_r} \text{ for some } r \text{ with } \mu < \kappa_r,$$

because every V_{κ_r} is elementary in V_λ. Let $\mu > \kappa_i$ belong to $(C^{(n)})^{V_\lambda}$, and choose $r \geq i$ with $\mu + \omega < \kappa_r$ and $V_\mu \prec_{\Sigma_n} V_{\kappa_r}$. Choose $s > r$ and $t > s$. The set-sized restriction $e = j^t \restriction V_{\kappa_s} : V_{\kappa_s} \longrightarrow V_{\kappa_{s+t}}$ is elementary by Lemma 2.1, has rank below $\kappa_{s+t} + \omega$, and therefore belongs to V_λ. Put $k = j^i(e)$, where this denotes the image of the set e under j^i. Then $k \in V_\lambda$, $k : V_{\kappa_{s+i}} \longrightarrow V_{\kappa_{s+t+i}}$, $\mathrm{crit}(k) = \kappa_i$, and $k(\kappa_i) = \kappa_{t+i} > \mu$. Since $r \geq i$, we have $\kappa_r = j^i(\kappa_{r-i})$, and hence

$$k(\kappa_r) = j^i(e(\kappa_{r-i})) = \kappa_{r+t}.$$

Applying k to $V_\mu \prec_{\Sigma_n} V_{\kappa_r}$ gives

$$V_{k(\mu)} \prec_{\Sigma_n} V_{\kappa_{r+t}}.$$

As $V_{\kappa_{r+t}} \prec V_\lambda$, the displayed characterization shows that $k(\mu) \in (C^{(n)})^{V_\lambda}$. Because k has critical point κ_i, V_λ regards κ_i as a cardinal. Indeed, a surjection $f : \xi \to \kappa_i$ with $\xi < \kappa_i$ would be carried to a surjection from ξ onto $k(\kappa_i)$, although $k(f)(\zeta) = f(\zeta) < \kappa_i$ for every $\zeta < \xi$. The map $k \restriction V_\mu \in V_\lambda$ is therefore the embedding required by $E_n(\kappa_i)$ for the chosen μ. Thus every κ_i is $C^{(n)}$-extendible in V_λ. Since the critical sequence is cofinal in λ and n was arbitrary, $V_\lambda \models \mathsf{UE}$. Lemma 5.1 now gives $V_\lambda \models \mathsf{VP}$. □

Lemma 5.3. *If $m > n \geq 1$, then every $C^{(m)}$-extendible cardinal is $C^{(n)}$-extendible.*

Proof. It is enough to consider $m = n + 1$. Let κ be $C^{(n+1)}$-extendible, and let $\lambda > \kappa$ belong to $C^{(n)}$. Choose $\mu > \lambda$ in $C^{(n+1)}$ and an elementary embedding $j : V_\mu \longrightarrow V_\nu$ with $\mathrm{crit}(j) = \kappa$, $j(\kappa) > \mu$, and $\nu \in C^{(n+1)}$. Because $\mu \in C^{(n+1)}$, we have $V_\mu \models \lambda \in C^{(n)}$. Elementarity therefore gives

$$V_\nu \models j(\lambda) \in C^{(n)}.$$

Since $\nu \in C^{(n+1)}$, $j(\lambda) \in C^{(n)}$ also holds in V. For $n = 1$, this follows from the Δ_2 definition of $C^{(1)}$, while for $n \geq 2$ it follows from the usual Π_n definition of $C^{(n)}$. Thus $j(\lambda) \in C^{(n)}$, and $j \restriction V_\lambda : V_\lambda \longrightarrow V_{j(\lambda)}$ witnesses the required $C^{(n)}$-extendibility of κ. □

Proposition 5.4.

$$\mathrm{Con}(\mathsf{ZF} + \mathsf{VP}) \quad \textit{implies} \quad \mathrm{Con}(\mathsf{ZF} + \mathsf{VP} + \mathsf{UE} + \text{“}\textit{there are no rank-Berkeley cardinals}''\text{)}.$$

Proof. Put $T = \mathsf{ZF} + \mathsf{VP}$. This is a recursively axiomatized theory: the instances of the parameterized Vopěnka scheme are generated effectively from formula codes. The assertion RB that a rank-Berkeley cardinal exists is a single first-order sentence, since elementarity between set ranks is expressed by their canonical satisfaction relations. The proof of Lemma 5.2 can be formalized in T, uniformly in formula codes. Hence T proves $\mathsf{RB} \longrightarrow \mathrm{Con}(T)$. If $T + \neg\mathsf{RB}$ were inconsistent, the deduction theorem would give $T \vdash \mathsf{RB}$ and hence $T \vdash \mathrm{Con}(T)$, contrary to Gödel's second incompleteness theorem when T is consistent. Thus $T + \neg\mathsf{RB}$ is consistent.

Now work in a model of $T + \neg\mathsf{RB}$. Fix $n \geq 2$. The scheme VP includes $\mathsf{VP}(\Pi_{n+1})$. Mohammd proves that if $\mathsf{VP}(\Pi_{n+1})$ holds while the $C^{(n)}$-extendible cardinals are bounded, then rank-Berkeley cardinals are unbounded [16, Proposition 6.5 and Theorem 6.7]. Thus the $C^{(n)}$-extendible cardinals are unbounded. For $n = 1$, boundedness of the $C^{(1)}$-extendibles implies boundedness of the $C^{(2)}$-extendibles by Lemma 5.3; the $n = 2$ case therefore gives unboundedly many $C^{(1)}$-extendibles as well. Hence UE holds. □

Local restoration of Choice. The next definitions record finite bounds on the Lévy complexity needed for reflection. Fix a Gödel numbering of formulas, a primitive-recursive prenex-normal-form operation, and a primitive-recursive function $\ell(e)$ bounding the Lévy complexity of the formula coded by e and of its negation. Let $\mathcal{D}$ be a finite list of formulas defining additional predicates, and write $e^{\mathcal{D}}$ for the result of eliminating those predicates. Let Σ be a finite set of codes for formulas whose free variables occur among the fixed ordered tuple

$$(\eta, \delta, \theta, \rho, \alpha, x).$$

Let $\Sigma^{\pm}$ be the closure of Σ under subformulas and negation. Its associated complexity bound is

$$c_{\mathcal{D}}(\Sigma) = 2 + \max\Big(\{0\} \cup \{\ell(e^{\mathcal{D}}) : e \in \Sigma^{\pm}\}\Big).$$

Thus two ranks in $C^{(c_{\mathcal{D}}(\Sigma))}$ agree on all formulas in Σ after interpreting the predicates defined by $\mathcal{D}$.

Let $\chi_{\mathcal{D},\Sigma}(\eta, \delta, \theta, \rho, \alpha, x)$ be the first-order formula

$$\exists\bar{\delta}\,\exists\bar{\theta}\,\exists\bar{\rho}\,\exists\bar{\alpha}\,\exists\bar{x}\,\exists J\ \Phi_{\mathcal{D},\Sigma},$$

where $\Phi_{\mathcal{D},\Sigma}$ is the conjunction of

$$\eta < \bar{\delta} < \bar{\theta} < \bar{\theta} + \omega < \bar{\rho} < \delta < \theta < \theta + \omega < \rho, \qquad \bar{\alpha} < \bar{\theta}, \qquad \bar{x} \in V_{\bar{\rho}},$$

the assertion that $J : V_{\bar{\rho}} \to V_{\rho}$ is elementary with respect to the canonical satisfaction relations, the equations

$$\mathrm{crit}(J) = \bar{\delta}, \quad J(\bar{\delta}) = \delta, \quad J(\bar{\theta}) = \theta, \quad J(\bar{\alpha}) = \alpha, \quad J(\bar{x}) = x,$$

the assertion $\bar{\rho} \in C^{(c_{\mathcal{D}}(\Sigma))}$, and the literal finite conjunction

$$\bigwedge_{e \in \Sigma} e^{\mathcal{D}}[\eta, \bar{\delta}, \bar{\theta}, \bar{\rho}, \bar{\alpha}, \bar{x}],$$

where the brackets denote primitive-recursive, capture-avoiding simultaneous substitution for the fixed tuple of free variables. Predicate elimination, simultaneous substitution, finite conjunction, rank membership, and set-coded satisfaction are effective operations. Hence the code of $\chi_{\mathcal{D},\Sigma}$ is primitive recursive in $\mathcal{D}$ and Σ. We use the following second complexity bound:

$$s_{\mathcal{D}}(\Sigma) = 2 + \max\{c_{\mathcal{D}}(\Sigma), \ell(\ulcorner \chi_{\mathcal{D},\Sigma} \urcorner)\}.$$

We write $V_\gamma \prec_{\Sigma_1^*} V$ for Spoerl's strengthened form of Σ_1-elementarity [19, Definition 20]. We use Woodin's choiceless definition of supercompactness: a cardinal δ is supercompact if, whenever $\gamma > \delta$, $V_\gamma \prec_{\Sigma_1^*} V$, and $a \in V_\gamma$, there are $\bar{\gamma} < \delta$, $\bar{\delta}$, $\bar{a} \in V_{\bar{\gamma}}$, and an elementary embedding $j : V_{\bar{\gamma}+1} \to V_{\gamma+1}$ such that $\mathrm{crit}(j) = \bar{\delta}$, $j(\bar{\delta}) = \delta$, $j(\bar{a}) = a$, and $V_{\bar{\gamma}} \prec_{\Sigma_1^*} V$ [19, Definition 21].

We use Woodin's local forcing for restoring Choice, with the completed-prefix convention and restricted lifting formulation proved in [23, Section 3]. Spoerl gives a detailed exposition [19, Definition 43, Theorem 44, and Lemma 45], and Woodin's original construction appears in [22, pp. 323–324]. In Spoerl's presentation, the construction assigns a cardinal κ_β to each stage β.

We call the cardinal through which the iteration has been defined its *endpoint*. A stage γ is *marked* if $\kappa_\gamma = \gamma$.

For an ordinal ξ, let DC_ξ denote dependent choice for histories of length ξ. Fix the presentation from [23, Section 3], and let a_η code its completed prefix through η: the carriers, orders, tops, projections, inclusions, prefix-replacement maps and cutoff table, including the seed. At a marked η, $a_\eta \in V_{\eta+\omega}$.

Theorem 5.5 (Woodin)**.** *Assume that* AC *fails, let* μ *be least such that* DC_μ *fails, and let* $\Lambda > \mu$ *be supercompact in Woodin's choiceless sense. There is a weakly homogeneous forcing* $\mathbb{Q}_\Lambda$*, uniformly first-order definable over* V_Λ*, such that whenever* $G \subseteq \mathbb{Q}_\Lambda$ *is generic:*

(1) $(V[G])_\Lambda \models \mathsf{ZFC}$*;*
(2) *every marked* $\gamma \leq \Lambda$ *satisfies*

$$V_\gamma[G \cap \mathbb{Q}_\gamma] = (V[G])_\gamma;$$

$\mathbb{Q}_\Lambda$ *factors over* $\mathbb{Q}_\gamma$ *with the canonical projection, and every supercompact* $\gamma \leq \Lambda$ *is marked;*
(3) *suppose* $J : V_{\bar\gamma+\omega} \to V_{\gamma+\omega}$ *is elementary,* $J(\bar\gamma) = \gamma < \Lambda$*,* $\mathrm{crit}(J) = \bar\delta < \bar\gamma < J(\bar\delta) = \delta$*,* γ *is marked, and* $J(a_{\bar\gamma}) = a_\gamma$*. Then* $\bar\gamma$ *is marked and there is an elementary* $e : (V[G])_{\bar\gamma} \longrightarrow (V[G])_\gamma$ *agreeing with* J *on ground elements of* $V_{\bar\gamma}$*. Its graph has rank below* $\gamma + \omega < \Lambda$*.*

Let $\mathcal{D}_{\mathrm{W}}$ be a fixed finite list of formulas defining this presentation, its complete codes a_η, and marked stages.

For each γ, let $\mathrm{Def}(V_\gamma)$ be the collection of subsets of V_γ definable over $(V_\gamma, \in)$ with parameters from V_γ.

Lemma 5.6. *If* γ *is supercompact in Woodin's choiceless sense, then* $V_\gamma \models \mathsf{ZF}$*.*

Proof. By [19, Definition 19 and the proof of Theorem 44], no map $V_\beta \to \gamma$ with $\beta < \gamma$ has unbounded range. If $a \in V_\gamma$, choose $\beta < \gamma$ with $a \in V_\beta$; any cofinal map $a \to \gamma$ would extend by a constant value to V_β. Lemma 3.1 now gives $V_\gamma \models \mathsf{ZF}$. □

Lemma 5.7. *For every* $k \geq 1$ *there is an effectively obtained first-order formula* $\vartheta_k(p, e, \vec\tau)$*, independent of the endpoint, with the following property. If Woodin's iteration is defined through a supercompact cardinal* γ*, then* ϑ_k *defines over* V_γ *the forcing relation for* $\mathbb{Q}_\gamma$ *for formulas of Lévy complexity at most* k*, and the truth lemma holds for prefix generics induced by an ambient generic for a later stage, including the endpoint itself.*

Proof. Use the completed presentation of [23, Section 3.1]. Every completed prefix through $\eta < \gamma$, with its order and projection maps, belongs to V_γ. The fixed defining dictionary computes these prefixes at the supercompact endpoint.

By the completed-stage construction, a marked inaccessible endpoint is a direct-limit stage and $\sup_{\alpha<\gamma} \kappa_\alpha = \gamma$. Let $e_{\alpha,\gamma}$ be the canonical iteration embedding and set

$$P_\alpha = e_{\alpha,\gamma}\text{“}\mathbb{Q}_\alpha.$$

Replacement in V_γ makes each P_α a set. Under the completed-prefix factorization, $e_{\alpha,\gamma}$ is the first-coordinate embedding, and reductions are obtained by strengthening that coordinate while leaving the tail unchanged. Thus the P_α form a definable increasing sequence of complete subforcings with $\mathbb{Q}_\gamma = \bigcup_{\alpha<\gamma} P_\alpha$. Lemma 5.6 gives $V_\gamma \models \mathsf{ZF}$. Compute bounded forcing at a prefix containing the names; completeness makes this independent of the prefix. Extend by negation, conjunction and the existential density clause over names in V_γ, as in [23, Section 3.3]. Such names represent exactly $(V[G])_\gamma$. The decision and witness classes definable over V_γ are externally ground sets, so the induced generic meets them and the truth lemma follows.

For fixed standard k, apply this translation to a partial satisfaction formula for level k, including both polarities. This gives ϑ_k, uniformly in γ; assign a fixed value to invalid codes. □

For $k \geq 1$, fix ϑ_k as in Lemma 5.7 and put $r_k = c_{\mathcal{D}_{\mathrm{W}}}(\{\vartheta_k\})$.

Lemma 5.8. *Suppose $\theta < \Lambda$ are supercompact in Woodin's choiceless sense, Woodin's iteration is defined through Λ, and $\theta, \Lambda \in C^{(r_k)}$. If $G \subseteq \mathbb{Q}_\Lambda$ is generic, then*

$$(V[G])_\theta \prec_{\Sigma_k} (V[G])_\Lambda.$$

Proof. Both endpoints are marked by Theorem 5.5. Let G_θ be induced by the complete embedding of $\mathbb{Q}_\theta$ into $\mathbb{Q}_\Lambda$; in the nested presentation, $G_\theta = G \cap \mathbb{Q}_\theta$. Every dense subclass of $\mathbb{Q}_\theta$ definable over V_θ with set parameters is externally a set, since V_θ and its satisfaction relation are sets. Completeness therefore gives that G_θ is generic over $(V_\theta, \mathrm{Def}(V_\theta))$. Since θ is marked,

$$V_\theta[G_\theta] = (V[G])_\theta.$$

Thus every $x \in (V[G])_\theta$ is τ^{G_θ} for some $\mathbb{Q}_\theta$-name $\tau \in V_\theta$.

Since $\theta, \Lambda \in C^{(r_k)}$, predicate elimination for $\mathcal{D}_{\mathrm{W}}$ gives, for every formula φ of complexity at most k, every tuple of $\mathbb{Q}_\theta$-names $\vec{\tau} \in V_\theta$, and every $q \in \mathbb{Q}_\theta$,

$$q \Vdash_{\mathbb{Q}_\theta} \varphi(\vec{\tau}) \quad \text{if and only if} \quad q \Vdash_{\mathbb{Q}_\Lambda} \varphi(\vec{\tau}). \tag{5}$$

The same equivalence holds for $\neg\varphi$ by the closure under negation in the definition of $c_{\mathcal{D}_{\mathrm{W}}}$.

Suppose first that $(V[G])_\Lambda \models \varphi(\vec{\tau}^{G_\theta})$. By Lemma 5.7, choose $p \in G$ with $p \Vdash_{\mathbb{Q}_\Lambda} \varphi(\vec{\tau})$. The dense set used below is

$$D = \{q \leq p \restriction \theta : q \in \mathbb{Q}_\theta \text{ and } q \Vdash_{\mathbb{Q}_\theta} \varphi(\vec{\tau})\}.$$

We show that D is dense below $p \restriction \theta$. Otherwise, some $q_0 \leq p \restriction \theta$ has no extension in D. The negation clause then gives $q_0 \Vdash_{\mathbb{Q}_\theta} \neg\varphi(\vec{\tau})$, and (5) gives $q_0 \Vdash_{\mathbb{Q}_\Lambda} \neg\varphi(\vec{\tau})$. The reduction property gives a common extension of p and q_0, contradicting the two forcing statements. The class $D \cup \{r \in \mathbb{Q}_\theta : r \perp p \restriction \theta\}$ is definable and dense in $\mathbb{Q}_\theta$. Genericity and $p \restriction \theta \in G_\theta$ therefore give $D \cap G_\theta \neq \varnothing$. For $q \in D \cap G_\theta$, forcing soundness from Lemma 5.7 gives $(V[G])_\theta \models \varphi(\vec{\tau}^{G_\theta})$.

Conversely, if $(V[G])_\theta \models \varphi(\vec{\tau}^{G_\theta})$, Lemma 5.7 gives $q \in G_\theta$ such that $q \Vdash_{\mathbb{Q}_\theta} \varphi(\vec{\tau})$. Equation (5) transfers this assertion to $\mathbb{Q}_\Lambda$, and forcing soundness gives $(V[G])_\Lambda \models \varphi(\vec{\tau}^{G_\theta})$. This proves the required Σ_k-elementarity. □

Lemma 5.9. *Let $S \geq s_{\mathcal{D}}(\Sigma)$, suppose $E_S(\delta)$, and let $\eta < \delta < \theta < \theta + \omega < \rho$, with $\rho \in C^{(c_{\mathcal{D}}(\Sigma))}$. If $\alpha < \theta$, $x \in V_\rho$, and*

$$V \models e^{\mathcal{D}}[\eta, \delta, \theta, \rho, \alpha, x] \quad \textit{for every } e \in \Sigma,$$

then $\chi_{\mathcal{D},\Sigma}(\eta, \delta, \theta, \rho, \alpha, x)$ holds.

Proof. Choose $\mu \in C^{(S)}$ above $\rho + \omega$ and let $k : V_\mu \to V_\nu$ witness $E_S(\delta)$. Thus $\nu \in C^{(S)}$, $\mathrm{crit}(k) = \delta$, and $k(\delta) > \mu > \rho$. The graph of $k \restriction V_\rho$ has rank below $k(\rho) + \omega = k(\rho + \omega) < \nu$, and Lemma 2.1 shows that this restriction is elementary from V_ρ to $V_{k(\rho)}$. Use the following tuple:

$$(\bar{\delta}, \bar{\theta}, \bar{\rho}, \bar{\alpha}, \bar{x}, J) = (\delta, \theta, \rho, \alpha, x, k \restriction V_\rho).$$

Thus $V_\nu \models \chi_{\mathcal{D},\Sigma}(\eta, k(\delta), k(\theta), k(\rho), k(\alpha), k(x))$. Here $\rho < k(\delta)$ and $k(\eta) = \eta$. By assumption, the original tuple satisfies the conjunction over Σ, and the $C^{(S)}$-correctness of V_ν makes the same conjunction true there. Elementarity of k pulls the assertion back to V_μ. Since $\mu \in C^{(S)}$ and the formula and its negation have Lévy complexity below S, the reflected assertion is true in V. □

Lemma 5.10. *Every E_s-cardinal is Σ_{s+2}-correct. Moreover, there is a fixed $d_{\mathrm{W}} < \omega$ such that every E_s-cardinal is supercompact in Woodin's sense whenever $s \geq d_{\mathrm{W}}$.*

Proof. An E_s-cardinal is s-choiceless extendible in the sense of Mohammd: the ordinary witness has critical point $\delta > \xi$ for every prescribed $\xi < \delta$. Its Σ_{s+2}-correctness is therefore [16, Proposition 2.2].

Fix formulas $\sigma_0(\xi)$ and $\sigma_1(\delta, \gamma, a)$ expressing $V_\xi \prec_{\Sigma_1^*} V$ and the small-embedding clause in Spoerl's Definition 21, and choose d_{W} above the Lévy complexities of these formulas and their negations. Let $s \geq d_{\mathrm{W}}$ and assume $E_s(\delta)$. Given $\gamma > \delta$ with $V_\gamma \prec_{\Sigma_1^*} V$ and $a \in V_\gamma$, choose an E_s-witness $k : V_\mu \to V_\nu$ with $\mu \in C^{(s)}$ above $\gamma + \omega$. By Lemma 2.1, $k \restriction V_{\gamma+1}$ witnesses in V_ν the required assertion for $k(\gamma), k(\delta), k(a)$. Elementarity reflects this to V_μ, and the choice of d_{W} makes the reflected map and the Σ_1^*-correctness of its source absolute to V. Thus δ is supercompact in Woodin's sense. □

We apply Woodin's forcing at an E_s-cardinal for large enough s and use Lemmas 5.9 and 5.10 to obtain a ZFC rank model satisfying any given finite subset of $\mathsf{ZFC} + \mathsf{VP}$.

Theorem 5.11. *For every $N \geq 1$ there is an effectively determined integer t_N such that the following holds. Suppose $V \models \mathsf{ZF} + \mathsf{UE} + \neg\mathsf{AC}$, and let μ be least such that DC_μ fails. Let $\Lambda > \mu$ be an E_{t_N}-cardinal, and let $G \subseteq \mathbb{Q}_\Lambda$ be generic for Woodin's forcing. Then $M = (V[G])_\Lambda$ satisfies ZFC and, for every $1 \leq n \leq N$, has a proper class of $C^{(n)}$-extendible cardinals.*

Proof. For $\gamma \leq \Lambda$, write $M_\gamma = (V[G])_\gamma$ for the rank-γ part of M. Let $k = N + 1$ and put $r = r_k$. Let Ξ_N be the finite set of formulas asserting that a stage θ belongs to $C^{(r)}$, is supercompact in Woodin's sense, and that $x = a_\theta$ is the complete code defined by $\mathcal{D}_{\mathrm{W}}$. The two complexity bounds needed below are

$$c_N = c_{\mathcal{D}_{\mathrm{W}}}(\Xi_N), \qquad s_N = \max\{r, d_{\mathrm{W}}, s_{\mathcal{D}_{\mathrm{W}}}(\Xi_N)\}.$$

Let t_N be four more than the maximum of s_N and the Lévy complexity bounds, for both polarities after predicate elimination, of $E_{s_N}(\xi)$, the assertion that the E_{s_N}-cardinals are unbounded, membership in $C^{(c_N)}$, and the assertion that $C^{(c_N)}$ is unbounded. Once the coding is fixed, t_N is primitive recursive in N.

By Lemma 5.10, $\Lambda \in C^{(t_N+2)}$ and Λ is supercompact in Woodin's sense. Theorem 5.5 gives $M \models \mathsf{ZFC}$. Since UE asserts that the E_{s_N}-cardinals are unbounded, the choice of t_N implies that they are unbounded below Λ; the same argument applies to $C^{(c_N)} \cap \Lambda$.

For each $1 \leq n \leq N$, the Bagaria–Poveda characterization says that, in M, a cardinal δ is $C^{(n)}$-extendible if and only if, for a proper class of $\theta \in (C^{(n+1)})^M$ and every $\alpha < \theta$, there are $\bar{\delta} < \bar{\theta} < \delta$, $\bar{\alpha} < \bar{\theta}$, and an elementary embedding $e : M_{\bar{\theta}} \to M_\theta$ such that $\bar{\theta} \in (C^{(n+1)})^M$, $\mathrm{crit}(e) = \bar{\delta}$, $e(\bar{\delta}) = \delta$, and $e(\bar{\alpha}) = \alpha$ [2, Corollary 2.7].

Fix $\eta < \Lambda$ above μ, and choose an E_{s_N}-cardinal $\delta > \eta$. Let $\theta > \delta$ be another such cardinal and let $\alpha < \theta$. Choose $\rho \in C^{(c_N)} \cap \Lambda$ above $\theta + \omega$, and put $x = a_\theta \in V_{\theta+\omega} \subseteq V_\rho$. Lemma 5.10 and Woodin's recursion show that these objects satisfy Ξ_N. Lemma 5.9 gives

$$\eta < \bar{\delta} < \bar{\theta} < \bar{\theta} + \omega < \bar{\rho} < \delta$$

and $J_0 : V_{\bar{\rho}} \to V_\rho$ such that

$$\mathrm{crit}(J_0) = \bar{\delta}, \quad J_0(\bar{\delta}) = \delta, \quad J_0(\bar{\theta}) = \theta, \quad J_0(\bar{\alpha}) = \alpha, \quad J_0(\bar{x}) = a_\theta.$$

The reflected instances of Ξ_N give $\bar{\theta} \in C^{(r)}$, show that $\bar{\theta}$ is supercompact in Woodin's sense, and identify $\bar{x} = a_{\bar{\theta}}$.

Set $J = J_0 \restriction V_{\bar{\theta}+\omega}$. Lemma 2.1 shows that $J : V_{\bar{\theta}+\omega} \longrightarrow V_{\theta+\omega}$ is elementary.

Since $a_{\bar{\theta}} \in V_{\bar{\theta}+\omega}$, the restriction retains $J(a_{\bar{\theta}}) = a_\theta$. Theorem 5.5(3) therefore gives directly an elementary $e : M_{\bar{\theta}} \to M_\theta$ with critical point $\bar{\delta}$, $e(\bar{\delta}) = \delta$, and $e(\bar{\alpha}) = \alpha$. The last equality follows from agreement with J on ground ordinals. Its graph has rank below $\theta + \omega < \Lambda$, so $e \in M$. The ranks $\bar{\theta}$, θ, and Λ are marked and belong to $C^{(r)}$.

Lemma 5.8, applied twice, gives

$$M_{\bar{\theta}} \prec_{\Sigma_{N+1}} M \quad \text{and} \quad M_\theta \prec_{\Sigma_{N+1}} M.$$

Consequently $\bar{\theta}, \theta \in (C^{(n+1)})^M$ for every $n \leq N$. As in the proof of Lemma 5.2, $M_{\bar{\theta}}$ regards the critical point $\bar{\delta}$ as a cardinal, so M_θ regards $\delta = e(\bar{\delta})$ as a cardinal. Any contrary witness in M already belongs to M_θ, so δ is a cardinal in M.

For the fixed δ, the ordinal θ can be chosen arbitrarily high below Λ, and $\alpha < \theta$ was arbitrary, so the characterization shows that δ is $C^{(n)}$-extendible for every $n \leq N$. The cardinal δ was arbitrary above η, and η was arbitrary above μ; hence these cardinals are unbounded below $\Lambda = \mathrm{Ord}^M$, so each class is proper in M. □

Lemma 5.12. *For every finite set F_0 of instances of the arbitrary-language Vopěnka scheme, there is an effectively determined $N \geq 1$ such that*

$$\mathsf{ZF} + \text{“the } E_N\text{-cardinals are unbounded”} \vdash F_0.$$

Proof. For each class formula, the proof of Lemma 5.1 uses a sufficient level computed from syntactic bounds for that formula and the fixed definitions of structure codes, syntax and satisfaction. Include both polarities in the bound. These bounds are independent of the language and parameter values. Take a common bound for the finitely many instances in F_0; see [23, Section 2]. □

Theorem 5.13 (Woodin)**.**

$$\mathrm{Con}(\mathsf{ZF}+\mathsf{VP}) \quad \textit{if and only if} \quad \mathrm{Con}(\mathsf{ZFC}+\mathsf{VP}).$$

Proof. By Proposition 5.4, take a countable model

$$\mathfrak{M} \models \mathsf{ZF}+\mathsf{VP}+\mathsf{UE}+\text{“there are no rank-Berkeley cardinals”}.$$

Let F be a finite subset of the first-order theory $\mathsf{ZFC}+\mathsf{VP}$. If $\mathfrak{M} \models \mathsf{AC}$, then $\mathfrak{M} \models F$.

Assume that $\mathfrak{M} \models \neg\mathsf{AC}$. Choose $N \geq 1$ for the instances of VP in F by Lemma 5.12, choose t_N as in Theorem 5.11, and choose in $\mathfrak{M}$ a sufficiently large E_{t_N}-cardinal Λ. Let $H \subseteq (\mathbb{Q}_\Lambda)^{\mathfrak{M}}$ be $\mathfrak{M}$-generic and put $W = (\mathfrak{M}[H])_\Lambda$. For a possibly ill-founded $\mathfrak{M}$, use its internal names modulo generic forced equality, with membership given by generic forced membership. The truth lemma holds for each standard formula. Theorem 5.11 gives $W \models \mathsf{ZFC}$ with unbounded E_N-cardinals, so Lemma 5.12 gives $W \models F$.

Every finite subset of $\mathsf{ZFC}+\mathsf{VP}$ is therefore satisfiable. By first-order compactness, $\mathsf{ZFC}+\mathsf{VP}$ has a model. The converse implication follows by forgetting Choice. □

This proves Theorem B.

Remark 5.14. For a finite set of axioms F, the rank extension used in the proof may depend on F. Compactness therefore proves equiconsistency, but does not give a class-generic $\mathsf{ZFC}+\mathsf{VP}$ extension of the original model. Question 9.2 asks whether every countable model admits one.

6. Independence of Choice principles

Theorem 3.3 applies to the classical symmetric constructions in this section and the next two; Theorem 5.13 gives a ground model of $\mathsf{ZFC}+\mathsf{VP}$ when one is required.

Corollary 6.1.

$$\mathrm{Con}(\mathsf{ZF}+\mathsf{VP}) \quad \textit{implies} \quad \mathrm{Con}(\mathsf{ZF}+\mathsf{VP}+\neg\mathsf{AC}+\neg\mathsf{DC}).$$

Consequently AC *is independent of* $\mathsf{ZF}+\mathsf{VP}$*, relative to* $\mathrm{Con}(\mathsf{ZF}+\mathsf{VP})$.

Proof. Let $\mathfrak{M}$ be a countable model of $\mathsf{ZF}+\mathsf{VP}$, and form its basic Cohen symmetric extension. The canonical set $A = \{x_n : n < \omega\}$ of Cohen reals is infinite and Dedekind-finite [12, Section 5.3]. Hence AC and DC fail. Theorem 3.3 preserves VP, while Theorem 5.13 gives the opposite consistency implication for AC. □

Lemma 6.2. *Assume* $\mathsf{ZFC}+\mathsf{VP}$*. There is a set-sized symmetric extension satisfying*

$$\mathsf{ZF}+\mathsf{VP}+\mathsf{DC}+\neg\mathsf{AC}.$$

Proof. Let $\mathbb{P}_1$ be the forcing of countable partial functions from $\omega_1 \times \omega_1$ to 2, ordered by reverse inclusion. Let the full permutation group of ω_1 act on the first coordinate, and let the normal filter be generated by pointwise stabilizers of countable subsets of ω_1. The forcing is σ-closed and the filter is σ-complete, so the symmetric extension satisfies DC by Karagila's preservation theorem [14, Lemma 3.1]. Theorem 3.3 gives $\mathsf{ZF}+\mathsf{VP}$.

For $\xi < \omega_1$, let a_ξ be the ξth generic subset of ω_1 and put $A = \{a_\xi : \xi < \omega_1\}$. The name for A is invariant, while the name for a_ξ has support $\{\xi\}$, so A belongs to the symmetric extension. Suppose that p forces a hereditarily symmetric name $\dot{f}$, supported by a countable $E \subseteq \omega_1$, to enumerate A. Enlarge E to contain the first-coordinate support of p. Choose $\xi \notin E$, $q \leq p$, and an ordinal γ such that

$$q \Vdash_{\mathrm{sym}} \dot{f}(\check{\gamma}) = \dot{a}_\xi.$$

Choose ζ outside E and outside the countable set of first coordinates occurring in q, and let π transpose ξ and ζ. Then π fixes $\dot{f}$ and p, and

$$\pi q \Vdash_{\mathrm{sym}} \dot{f}(\check{\gamma}) = \dot{a}_{\zeta}.$$

Since q and πq are compatible, choose a common extension. It forces $\dot{a}_{\xi} \neq \dot{a}_{\zeta}$ while forcing both names to equal $\dot{f}(\check{\gamma})$, a contradiction. Thus A is not well-orderable, and AC fails. □

Theorem 6.3.

$$\mathrm{Con}(\mathsf{ZF}+\mathsf{VP}) \quad \textit{implies} \quad \mathrm{Con}(\mathsf{ZF}+\mathsf{VP}+\mathsf{DC}+\neg\mathsf{AC}).$$

Proof. Let $\mathfrak{M} \models \mathsf{ZF}+\mathsf{VP}$ be countable. If $\mathfrak{M} \models \mathsf{AC}$, apply Lemma 6.2. If $\mathfrak{M} \models \neg\mathsf{AC}+\neg\mathsf{DC}$, choose a singular weakly LS cardinal κ by Corollary 4.4. Usuba's forcing $\mathrm{Col}(V_\kappa)$ forces DC [21, Proposition 3.6], while Theorem 3.3 preserves VP; if the extension satisfies AC, apply Lemma 6.2 once more. The remaining case already satisfies $\mathsf{DC}+\neg\mathsf{AC}$. □

Corollary 6.4. *Relative to* $\mathrm{Con}(\mathsf{ZF}+\mathsf{VP})$*, the axiom* DC *is independent of*

$$\mathsf{ZF}+\mathsf{VP}+\neg\mathsf{AC}.$$

Proof. Corollary 6.1 gives a model with $\neg\mathsf{DC}$, and Theorem 6.3 gives one with DC, from the same consistency hypothesis. □

Let RR denote the rigid relation principle: every set carries a binary relation with no nonidentity automorphism.

Corollary 6.5. *The following consistency implications hold:*

$$\mathrm{Con}(\mathsf{ZF}+\mathsf{VP}) \quad \textit{implies} \quad \mathrm{Con}(\mathsf{ZF}+\mathsf{VP}+\neg\mathsf{RR}),$$
$$\mathrm{Con}(\mathsf{ZF}+\mathsf{VP}) \quad \textit{implies} \quad \mathrm{Con}(\mathsf{ZF}+\mathsf{VP}+\mathsf{RR}+\neg\mathsf{AC}+\neg\mathsf{DC}).$$

Proof. Hamkins and Palumbo prove that every model of ZF has a set-sized symmetric extension in which RR fails, and that RR holds in the basic Cohen model [8, Theorems 5 and 6]. Apply Theorem 3.3 to the first construction over a model of $\mathsf{ZF}+\mathsf{VP}$. For the second implication, first use Theorem 5.13 to obtain a model of $\mathsf{ZFC}+\mathsf{VP}$, and then use the Cohen model from the proof of Corollary 6.1; it satisfies $\neg\mathsf{AC}+\neg\mathsf{DC}$, and Theorem 3.3 preserves VP. □

7. The Feferman–Lévy model

Theorem 7.1. *If* $\mathsf{ZF}+\mathsf{VP}$ *is consistent, then so is* $\mathsf{ZF}+\mathsf{VP}+\neg\mathsf{AC}+\neg\mathsf{DC}$ *together with the assertion that* $\mathbb{R}$ *is a countable union of countable sets.*

Proof. By Theorem 5.13, it suffices to start with $V \models \mathsf{ZFC}+\mathsf{VP}$. Let $\mathbb{P}_0$ be trivial, let N_0 consist of the canonical names for the ground-model reals, and choose an infinite regular cardinal $\kappa_1 > |N_0|$. Recursively, after $\kappa_1, \dots, \kappa_n$ have been chosen, define the finite-stage forcing by

$$\mathbb{P}_n = \prod_{1\le m\le n}^{\mathrm{fin}} \mathrm{Coll}(\omega, \kappa_m),$$

let N_n be the set of nice $\mathbb{P}_n$-names for reals, and choose an infinite regular cardinal $\kappa_{n+1} > \max\{\kappa_n, |N_n|\}$. Following the standard Feferman–Lévy construction [12, Theorem 10.6], use the full forcing

$$\mathbb{P} = \prod_{1\le m<\omega}^{\mathrm{fin}} \mathrm{Coll}(\omega, \kappa_m).$$

Let Γ consist of the permutations acting independently on the ω-coordinate of each column, and let $\mathcal{F}$ be generated by the subgroups H_n fixing the first n columns pointwise, with $H_0 = \Gamma$. Let G be generic and $W = \mathsf{HS}^G$. Proposition 2.2 and Theorem 3.3 give

$$W \models \mathsf{ZF}+\mathsf{VP}.$$

For $n < \omega$, let G_n be the restriction of G to columns $1, \dots, n$, with G_0 trivial. We use the notation

$$R_n = \mathbb{R}^{V[G_n]}.$$

If $x \in W$ is a real, $\dot{x}$ has support H_n, and $p \Vdash_{\mathrm{sym}} \check{k} \in \dot{x}$, the usual support argument gives

$$p \restriction n \Vdash_{\mathrm{sym}} \check{k} \in \dot{x}.$$

If $q \leq p \restriction n$ forced the opposite decision, choose $\pi \in H_n$ moving the finitely many coordinates of p above column n away from those of q. Then πp and q are compatible and π fixes $\dot{x}$, a contradiction. Therefore

$$x = \{k \in \omega : \exists q \in G_n \ (q \Vdash_{\mathbb{P}} \check{k} \in \dot{x})\},$$

and ordinary and symmetric forcing agree in this atomic formula for these hereditarily symmetric names. Hence $x \in V[G_n]$. Conversely, every $\mathbb{P}_n$-name is hereditarily fixed by H_n, and hence $V[G_n] \subseteq W$. Consequently,

$$\mathbb{R}^W = \bigcup_{n<\omega} R_n.$$

Regard the set N_n fixed above as a set of $\mathbb{P}$-names. The names used below are

$$\dot{R}_n = \{\langle \dot{y}, 1_{\mathbb{P}} \rangle : \dot{y} \in N_n\}, \qquad \dot{S} = \{\langle \langle \check{n}, \dot{R}_n \rangle^{\bullet}, 1_{\mathbb{P}} \rangle : n < \omega\}.$$

Every automorphism preserves the first n columns as a set and permutes N_n, so $\dot{R}_n$ and $\dot{S}$ are invariant. Their constituent names are supported by H_n, hence $\dot{S} \in \mathsf{HS}$ and

$$\dot{S}^G = \langle R_n : n < \omega \rangle.$$

Hence the entire sequence lies in W.

Finally, each R_n is countable in W. Fix $n < \omega$ and, in V, well-order N_n. Mapping each real in R_n to the least index of a nice name evaluating to it gives, in $V[G_n] \subseteq W$, an injection $e_n : R_n \longrightarrow |N_n|^V$, where $|N_n|^V < \kappa_{n+1}$. The next generic column adds a surjection $g_{n+1} : \omega \to \kappa_{n+1}$. Extending e_n^{-1} by a fixed value off $\mathrm{ran}(e_n)$ and composing with g_{n+1} gives a surjection from ω onto R_n. For $n = 0$, the filter G_0 is trivial, N_0 is the chosen set of canonical names, and g_1 is the first generic column.

Thus W contains the sequence $\langle R_n : n < \omega \rangle$ of countable sets with union $\mathbb{R}^W$. By Cantor's theorem, $\mathbb{R}^W$ is uncountable. Hence countable choice fails, and so does DC. □

8. The full Solovay model

Suppose for this section that $V \models \mathsf{ZFC} + \mathsf{VP}$. By the final clause of Lemma 3.2, fix a measurable cardinal κ and a nonprincipal κ-complete ultrafilter U on κ. Let $G \subseteq \mathrm{Coll}(\omega, <\kappa)$ be V-generic, put $R = (\omega^\omega)^{V[G]}$, and define

$$\mathcal{S}ol = \mathrm{HOD}^{V[G]}_{V \cup R}, \tag{6}$$

where individual elements of V and individual members of R are allowed as parameters. We use LM, BP, and PSP for the assertions that every set of reals is, respectively, Lebesgue measurable, has the Baire property, and has the perfect set property.

Lemma 8.1. *The model $\mathcal{S}ol$ satisfies $\mathsf{ZF} + \mathsf{DC} + \mathsf{LM} + \mathsf{BP} + \mathsf{PSP}$, and $\kappa = \omega_1^{\mathcal{S}ol}$.*

Proof. The model in (6) is the full Solovay model in the precise convention of [17, Definition 4 and the following paragraph]. Since a measurable cardinal is inaccessible in ZFC, Solovay's argument gives $\mathcal{S}ol \models \mathsf{ZF} + \mathsf{DC} + \mathsf{LM} + \mathsf{BP} + \mathsf{PSP}$ [18, III, Sections 1.3–1.11 and 2.4–2.11]. These arguments allow ground-set parameters and give the countable-sequence closure needed for the full hereditary model.

The collapse makes every $\alpha < \kappa$ countable in $\mathcal{S}ol$. Since κ is inaccessible, $\mathrm{Coll}(\omega, <\kappa)$ is κ-c.c. and preserves the uncountability of κ. Hence $\kappa = \omega_1^{\mathcal{S}ol}$. □

Lemma 8.2. *If $x \in \mathcal{S}ol$ is a set of ordinals, then $x \in V[G_\beta]$ for some $\beta < \kappa$, where $G_\beta = G \cap \mathrm{Coll}(\omega, <\beta)$. Moreover, if $\langle A_\xi : \xi < \delta \rangle \in \mathcal{S}ol$, where $\delta < \kappa$ and every $A_\xi \subseteq \kappa$, then the whole sequence belongs to one such bounded-stage extension.*

Proof. Let $x \in \mathcal{S}ol$ be a set of ordinals, and choose an ordinal θ with $x \subseteq \theta$. There are a formula φ, ordinal parameters $\vec{\gamma}$, a parameter $a \in V$, and a real $r \in V[G]$ such that

$$\alpha \in x \quad \text{if and only if} \quad V[G] \models \varphi(\alpha, \vec{\gamma}, a, r) \qquad (\alpha < \theta).$$

A nice name for r has support of size less than κ, and hence bounded support, so $r \in V[G_\beta]$ for some $\beta < \kappa$. Let $\mathbb{P}^\beta$ be the quotient forcing over $V[G_\beta]$. Since $\mathbb{P}^\beta$ is weakly homogeneous and its automorphisms fix the parameters,

$$x = \{\alpha < \theta : 1_{\mathbb{P}^\beta} \Vdash \varphi(\check{\alpha}, \check{\vec{\gamma}}, \check{a}, \check{r})\}.$$

The definability of the forcing relation and Separation in $V[G_\beta]$ therefore give $x \in V[G_\beta]$. This is the localization argument for the full Solovay model [17, Definition 4, Fact 5, and the paragraph following Fact 5].

For the final assertion, code the sequence by

$$C = \{\kappa \cdot \xi + \alpha : \xi < \delta,\ \alpha \in A_\xi\}.$$

Then $C \in \mathcal{S}ol$, so $C \in V[G_\beta]$ for some $\beta < \kappa$. Since

$$A_\xi = \{\alpha < \kappa : \kappa \cdot \xi + \alpha \in C\},$$

the sequence $\langle A_\xi : \xi < \delta \rangle$ belongs to $V[G_\beta]$. □

Lemma 8.3. *In $\mathcal{S}ol$, the set*

$$U^* = \{A \in \mathcal{P}(\kappa)^{\mathcal{S}ol} : (\exists Y \in U)\, Y \subseteq A\}$$

is a nonprincipal, internally κ-complete ultrafilter on κ.

Proof. Since $U \in V \subseteq \mathcal{S}ol$, Separation gives $U^* \in \mathcal{S}ol$. For $\beta < \kappa$, define

$$U_\beta = \{X \in \mathcal{P}(\kappa)^{V[G_\beta]} : (\exists Y \in U)\, Y \subseteq X\}.$$

The forcing $\mathrm{Coll}(\omega, {<}\beta)$ has ground-model cardinality less than κ, so [3, Theorem 2.5(2)] shows that U_β is a κ-complete ultrafilter. If $A \subseteq \kappa$ belongs to $\mathcal{S}ol$, choose $\beta < \kappa$ with $A \in V[G_\beta]$ by Lemma 8.2. Then $A \in U^*$ if and only if $A \in U_\beta$, so exactly one of A and $\kappa \setminus A$ belongs to U^*.

To verify internal κ-completeness, let $\delta < \kappa$ and $\langle A_\xi : \xi < \delta \rangle \in \mathcal{S}ol$, with every $A_\xi \in U^*$. By Lemma 8.2, the sequence $\langle A_\xi : \xi < \delta \rangle$ belongs to $V[G_\beta]$ for some $\beta < \kappa$. Since each $A_\xi \in U^*$, we have $A_\xi \in U_\beta$. The κ-completeness of U_β therefore gives

$$\bigcap_{\xi < \delta} A_\xi \in U_\beta.$$

The intersection belongs to $\mathcal{S}ol$, and its membership in U_β therefore implies that it belongs to U^*. Finally, no singleton belongs to U^* because U is nonprincipal. Thus U^* is nonprincipal and internally κ-complete. □

Lemma 8.4. *The model $\mathcal{S}ol$ is the symmetric extension arising from a set-sized symmetric system in V.*

Proof. Let $\mathbb{B}$ be the Boolean completion of $\mathrm{Coll}(\omega, {<}\kappa)$, let $H \subseteq \mathbb{B}$ be the induced generic, and define the Boolean class name below using the condition-first convention of [15]:

$$\dot{R}_{\mathrm{can}} = \{(b, \tau) : \tau \text{ is a } \mathbb{B}\text{-name and } 0 < b \leq \|\tau \in \check{\omega}^{\check{\omega}}\|_{\mathbb{B}}\}.$$

Then $\dot{R}_{\mathrm{can}}^H = R$. Equivariance of Boolean values gives $\pi \dot{R}_{\mathrm{can}} = \dot{R}_{\mathrm{can}}$ for every $\pi \in \mathrm{Aut}(\mathbb{B})$. Moreover, $\dot{R}_{\mathrm{can}}$ self-reflects in the sense of [15, Definitions 9.1–9.2]: for any fixed ground-model real r_0, both $\dot{R}_{\mathrm{can}}$ and $\check{r}_0$ have stabilizer $\mathrm{Aut}(\mathbb{B})$, and $1_{\mathbb{B}} \Vdash \check{r}_0 \in \dot{R}_{\mathrm{can}}$.

Consequently, by [15, Proposition 9.4], using same-extension comparison generics as in [23, Proposition 4.2], the symmetric extension arising from $\mathcal{O}(\mathbb{B}, \dot{R}_{\mathrm{can}})$ is exactly

$$\mathrm{HOD}_{V \cup R}^{V[H]} = \mathcal{S}ol,$$

using the displayed self-reflecting name. This symmetric system is set-sized in V: $\mathbb{B}$ and $\mathrm{Aut}(\mathbb{B})$ are sets, and its generated normal filter is a definable subset of $\mathcal{P}(\mathrm{Aut}(\mathbb{B}))$. □

Corollary 8.5. $\mathrm{Con}(\mathsf{ZF}+\mathsf{VP})$ *implies the consistency of*

$$\mathsf{ZF}+\mathsf{VP}+\mathsf{DC}+\mathsf{LM}+\mathsf{BP}+\mathsf{PSP}+\neg\mathsf{AC}+\text{“}\omega_1 \textit{ carries a nonprincipal } \omega_1\textit{-complete ultrafilter.}''$$

Proof. By Theorem 5.13, take the starting model to satisfy $\mathsf{ZFC}+\mathsf{VP}$. By Lemma 8.4 and Theorem 3.3, $\mathcal{S}ol \models \mathsf{VP}$. Lemma 8.1 gives $\mathsf{ZF}+\mathsf{DC}+\mathsf{LM}+\mathsf{BP}+\mathsf{PSP}$ and identifies κ with $\omega_1^{\mathcal{S}ol}$, while Lemma 8.3 gives the nonprincipal, internally κ-complete ultrafilter. The standard Solovay argument gives $\neg\mathsf{AC}$. □

9. Limits of Choice restoration

Question 9.2 asks whether every countable model of $\mathsf{ZF}+\mathsf{VP}$ has a class-generic extension satisfying $\mathsf{ZFC}+\mathsf{VP}$. Under the stated forcing-theorem hypotheses, Proposition 9.1 characterizes when a class forcing preserves $\mathsf{VP}(\Pi_m)$. Proposition 9.4 gives a positive answer under SVC. Corollary 9.6 rules out well-orderable set forcing, while Proposition 9.7 gives an $\aleph_1$-sized counterexample to the version without countability.

Proposition 9.1. *Let $(M,\mathcal{X}) \models \mathsf{GBC}$ be a countable transitive two-sorted model, where GBC denotes Gödel–Bernays set theory with Global Choice, and let $\mathbb{P} \in \mathcal{X}$ be a class forcing with a largest condition. Assume that the forcing theorem holds for first-order formulas, that the relevant forcing relations belong to $\mathcal{X}$, that a generic may be chosen through every condition, and that every generic extension has the same ordinals and satisfies ZFC. For standard $m \geq 1$, let $K_m(\kappa)$ mean that κ is supercompact if $m=1$, and $C^{(m-1)}$-extendible if $m \geq 2$. Then*

$$1_{\mathbb{P}} \Vdash \mathsf{VP}(\Pi_m) \quad \textit{if and only if} \quad \forall p \in \mathbb{P}\ \forall \eta \in \mathrm{Ord}^M\ \exists q \leq p\ \exists \kappa > \eta\ q \Vdash K_m(\check{\kappa}).$$

Here the left side means that every coded instance of $\mathsf{VP}(\Pi_m)$ is forced. Consequently, $\mathbb{P}$ forces the full first-order scheme exactly when the right side holds for every standard $m \geq 1$.

Proof. Over ZFC, $\mathsf{VP}(\Pi_m)$ implies unboundedly many K_m-cardinals [1, Theorem 4.3(2), Corollary 4.7, and the paragraph following Theorem 4.12]. Conversely, for a Π_m class choose such a cardinal κ above its language and parameters and a class member outside H_κ. Regarding the definition as Σ_{m+1}, [1, Theorems 4.5 and 4.11] gives a distinct reflected member in H_κ. Thus the equivalence holds with our language convention. Suppose first that $\mathsf{VP}(\Pi_m)$ is forced, and fix p and η. Choose a generic G containing p. Some $\kappa > \eta$ in $M[G]$ satisfies K_m, and the preservation of ordinals gives $\kappa \in M$. By the assumed truth lemma and directedness of G, some $q \leq p$ forces $K_m(\check{\kappa})$.

Conversely, for each η, the conditions forcing a K_m-cardinal above η are dense, so every generic extension satisfies $\mathsf{VP}(\Pi_m)$. For each instance, the hypotheses that a generic can be chosen through every condition and that the truth lemma holds imply that the conditions forcing it are dense. Hence $1_{\mathbb{P}}$ forces every instance of $\mathsf{VP}(\Pi_m)$. □

Question 9.2. For every countable first-order model $M \models \mathsf{ZF}+\mathsf{VP}$, do there exist a class part $\mathcal{X}$ over M, a class forcing $\mathbb{P} \in \mathcal{X}$, and an externally $(M,\mathcal{X})$-generic filter $G \subseteq \mathbb{P}$ such that the quotient-of-names extension satisfies $M[G] \models \mathsf{ZFC}+\mathsf{VP}$?

Question 9.3. For each standard $n \geq 1$, is it relatively consistent with $\mathsf{ZFC}+\mathsf{VP}(\Pi_{n+1})$ that a class forcing, definable in the language of set theory from set parameters, preserves ZFC and $\mathsf{VP}(\Pi_n)$ but destroys $\mathsf{VP}(\Pi_{n+1})$?

Two related results bear on Question 9.3. For $n \geq 2$, suitable Δ_2-definable class iterations preserve $\mathsf{VP}(\Pi_n)$ over models of $\mathsf{VP}(\Pi_{n+1})$ [2, Theorem 9.6]. On the other hand, Hamkins shows that the first-order scheme need not imply Vopěnka's principle for arbitrary classes [7, Theorem 11].

Blass introduced the axiom of small violations of Choice, SVC. We use his characterization that, over ZF, SVC is equivalent to the existence of a set forcing which forces AC [4]; see also [14, Section 3].

Proposition 9.4. *Let $M \models \mathsf{ZF}+\mathsf{VP}+\mathsf{SVC}$. There is a set forcing $\mathbb{P} \in M$ such that every M-generic extension by $\mathbb{P}$ satisfies $\mathsf{ZFC}+\mathsf{VP}$. In particular, if M is countable, Question 9.2 has a positive answer for M.*

Proof. By Blass's characterization, choose $\mathbb{P} \in M$ with $1_{\mathbb{P}} \Vdash \mathsf{AC}$. Set forcing preserves ZF, and the trivial-symmetry case of Theorem 3.3 preserves VP. Thus $1_{\mathbb{P}} \Vdash \mathsf{ZFC} + \mathsf{VP}$. □

Lemma 9.5. *Let $\mathbb{P}$ be a well-orderable set forcing, X a ground-model set, and suppose that for some ordinal κ and name $\dot{f}$,*

$$1_{\mathbb{P}} \Vdash \dot{f} : \check{\kappa} \twoheadrightarrow \check{X}.$$

Then X is well-orderable in the ground model.

Proof. Fix a well-order of $\mathbb{P} \times \kappa$. For $x \in X$, let $d(x)$ be the least pair (p, α) such that $p \Vdash \dot{f}(\check{\alpha}) = \check{x}$. Since $1_{\mathbb{P}}$ forces $\dot{f}$ to be surjective, Replacement yields a function $d : X \longrightarrow \mathbb{P} \times \kappa$. If $d(x) = d(y) = (p, \alpha)$, then p forces both $\dot{f}(\check{\alpha}) = \check{x}$ and $\dot{f}(\check{\alpha}) = \check{y}$, so $x = y$. Thus d is injective, and hence X is well-orderable. □

Corollary 9.6. *A well-orderable set forcing cannot restore Choice over a model of $\neg\mathsf{AC}$.*

Proof. Let a generic extension by a well-orderable $\mathbb{P}$ satisfy AC, and fix a ground-model set X. In the extension, choose a surjection $f : \kappa \twoheadrightarrow X$ from an ordinal κ. By the truth lemma, there are a condition $p \in \mathbb{P}$ and a name $\dot{f}$ such that $p \Vdash \dot{f} : \check{\kappa} \twoheadrightarrow \check{X}$. After restricting $\dot{f}$ below p, Lemma 9.5 applied to the well-orderable forcing $\mathbb{P} \restriction p$ shows that X is well-orderable in the ground model. Thus every ground-model set is well-orderable, and the ground model satisfies AC. □

Proposition 9.7. *Assume* $\mathrm{Con}(\mathsf{ZF} + \mathsf{VP})$. *There is a model N of cardinality $\aleph_1$ satisfying $\mathsf{ZF} + \mathsf{VP} + \neg\mathsf{AC}$ which has no proper generic extension satisfying ZF. Consequently, the analogue of Question 9.2 for all set-sized models is false.*

Proof. By Corollary 6.1, take a countable $M_0 \models \mathsf{ZF} + \mathsf{VP} + \neg\mathsf{AC}$. Enayat's theorem gives an elementary extension $M_0 \prec N$ of cardinality $\aleph_1$ having no proper end extension to a model of ZF [6, Theorem 5.18], using the membership-language existence theorem in [23, Theorem 5.1]. Elementarity gives $N \models \mathsf{ZF} + \mathsf{VP} + \neg\mathsf{AC}$. The atomic forcing recursion shows that old sets acquire no new members in a quotient-of-names extension. Hence any such extension that adds a set is both proper and an end extension. Therefore no such extension of N can satisfy ZF. □

Concluding remarks

We propose collaborations between mathematicians and contributors exploring set theory with LLM assistance to develop detailed arguments and understand their relation to existing work. Early-career researchers can be valuable partners, contributing to the proofs and drawing on their connections to specialists to clarify the methods and history.

Carefully checked drafts can support ongoing discussion of the arguments and their provenance. These exchanges can help identify and document results previously known only informally. We welcome questions, corrections, and pointers to earlier work concerning this paper. This feedback can inform subsequent revisions, with appropriate credit for both the original results and contributions to their written proofs.

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